clinical_analyses

```r
# Packages
#install.packages(c("dplyr", "lme4", "lmerTest", "purrr", "broom.mixed","rlang"))
```

```r
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.4.3

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(lme4)
```

```
## Warning: package 'lme4' was built under R version 4.4.3

## Loading required package: Matrix
```

```r
library(lmerTest)
```

```
## Warning: package 'lmerTest' was built under R version 4.4.3

##
## Attaching package: 'lmerTest'

## The following object is masked from 'package:lme4':
##
##     lmer

## The following object is masked from 'package:stats':
##
##     step
```

```r
library(purrr)
```

```
## Warning: package 'purrr' was built under R version 4.4.3
```

```r
library(broom.mixed)
```

```
## Warning: package 'broom.mixed' was built under R version 4.4.3
```

```r
library(rvest)
library(ggplot2)
library(emmeans)
```

```
## Welcome to emmeans.
## Caution: You lose important information if you filter this package's results.
## See '? untidy'
```

```r
library(dplyr)
library(tidyr)
```

```
##
## Attaching package: 'tidyr'

## The following objects are masked from 'package:Matrix':
##
##     expand, pack, unpack
```

```r
# Read data
real <- read.csv(
  "/Users/shamiraventurini/PycharmProjects/ICL-PILOT/phase2/ENNI/measures/severity_features_raw_real.cs
  check.names = FALSE
)

syn <- read.csv(
  "/Users/shamiraventurini/PycharmProjects/ICL-PILOT/phase2/generated/measures/severity_features_raw_sy
  check.names = FALSE
)

roster <- read.csv(
  "/Users/shamiraventurini/PycharmProjects/ICL-PILOT/configs/generation/four_year_old_frozen_roster.csv
  check.names = FALSE
)

# ------------------------------
# Prompt children used in each round
# Source of truth here = syn metadata
# ------------------------------
prompt_map <- syn %>%
  transmute(
    round_id = as.character(round_id),
    prompt_td_child_id = as.character(prompt_td_child_id),
```

```r
    prompt_sli_child_id = as.character(prompt_sli_child_id)
  ) %>%
  distinct() %>%
  pivot_longer(
    cols = c(prompt_td_child_id, prompt_sli_child_id),
    names_to = "prompt_group",
    values_to = "child_id"
  ) %>%
  filter(!is.na(child_id), child_id != "") %>%
  mutate(
    clinical_group = case_when(
      prompt_group == "prompt_td_child_id"  ~ "TD",
      prompt_group == "prompt_sli_child_id" ~ "SLI"
    )
  ) %>%
  select(round_id, clinical_group, child_id) %>%
  distinct()

# -----------------------------
# Standardize real pool
# -----------------------------
real_4yo <- real %>%
  mutate(
    File_ID = as.character(File_ID),
    Age_Month = as.numeric(`Age(Month)`),
    source = "real",
    pair_id = NA_character_,
    round_id = NA_character_,
    clinical_group = case_when(
      Group == "TD"  ~ "TD",
      Group == "SLI" ~ "SLI",
      TRUE ~ NA_character_
    ),
    comparison_target_group = clinical_group
  ) %>%
  filter(
    !is.na(clinical_group),
    Age_Month >= 48, Age_Month < 60
  )

# -----------------------------
# Standardize synthetic pool
# -----------------------------
syn_4yo <- syn %>%
  mutate(
    File_ID = as.character(File_ID),
    pair_id = as.character(pair_id),
    round_id = as.character(round_id),
    Age_Month = as.numeric(`Age(Month)`),
    source = "synthetic",
    clinical_group = case_when(
      comparison_target_group == "TD"  ~ "TD",
      comparison_target_group == "SLI" ~ "SLI",
```

```r
      TRUE ~ NA_character_
    )
  ) %>%
  filter(
    !is.na(clinical_group),
    Age_Month >= 48, Age_Month < 60
  )

# ------------------------------
# Build fold-stacked held-out evaluation dataframe
# For each round:
# - real children are included only if they were NOT prompts in that round
# - synthetic children come from that same round
# ------------------------------
rounds <- sort(unique(syn_4yo$round_id))

df_4pop <- bind_rows(lapply(rounds, function(r) {
  prompt_r <- prompt_map %>%
    filter(round_id == r)

  real_r <- real_4yo %>%
    mutate(round_id = r) %>%
    anti_join(
      prompt_r,
      by = c("File_ID" = "child_id", "clinical_group")
    )

  syn_r <- syn_4yo %>%
    filter(round_id == r)

  bind_rows(real_r, syn_r)
})) %>%
  filter(!is.na(Age_Month)) %>%
  mutate(
    source = factor(source, levels = c("real", "synthetic")),
    clinical_group = factor(clinical_group, levels = c("TD", "SLI")),
    population = factor(
      paste(source, clinical_group, sep = "_"),
      levels = c("real_TD", "synthetic_TD", "real_SLI", "synthetic_SLI")
    ),
    source_is_synthetic = if_else(source == "synthetic", 1L, 0L),
    cluster_id = if_else(
      source == "synthetic",
      paste0("synthetic_", pair_id),
      paste0("real_", File_ID)
    ),
    unit_id = cluster_id,
    Age_c = Age_Month - mean(Age_Month, na.rm = TRUE)
  )

# Optional subsets
df_td <- df_4pop %>% filter(clinical_group == "TD")
df_sli <- df_4pop %>% filter(clinical_group == "SLI")
```

```r
# Quick checks
table(df_4pop$source, df_4pop$clinical_group)
```

```
##
##              TD SLI
##   real      161  33
##   synthetic 110 110
```

```r
table(df_4pop$round_id, df_4pop$population)
```

```
##
##            real_TD synthetic_TD real_SLI synthetic_SLI
##   round_01      40           30        8            30
##   round_02      40           30        8            30
##   round_03      40           30        8            30
##   round_04      41           20        9            20
```

```r
summary(df_4pop$Age_Month)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   48.00   53.98   56.00   55.23   57.00   59.20
```

Classic Measures

mlu

```r
m0_mlu <- lmer(
  `MLU_Morphemes` ~ source_is_synthetic * clinical_group + factor(round_id) + (1 | cluster_id),
  data = df_4pop, REML = FALSE
) # do real vs syn and td vs sli differ? does the difference depend from clinical group?

m1_mlu <- lmer(
  `MLU_Morphemes` ~ source_is_synthetic * clinical_group + Age_c + factor(round_id) + (1 | cluster_id),
  data = df_4pop, REML = FALSE
) # do real vs syn and td vs sli differ? does the difference depend from clinical group? does mlu chang

m2_mlu <- lmer(
  `MLU_Morphemes` ~ source_is_synthetic * clinical_group + Age_c +
    source_is_synthetic:Age_c + clinical_group:Age_c + factor(round_id) +
    (1 | cluster_id),
  data = df_4pop, REML = FALSE
) # do real vs syn and td vs sli differ? does the difference depend from clinical group? does mlu chang

m3_mlu <- lmer(
  `MLU_Morphemes` ~ source_is_synthetic * clinical_group * Age_c + factor(round_id) + (1 | cluster_id),
  data = df_4pop, REML = FALSE
) # # do real vs syn and td vs sli differ? does the difference depend from clinical group? does mlu cha

anova(m0_mlu, m1_mlu, m2_mlu, m3_mlu)
```

```
## Data: df_4pop
## Models:
## m0_mlu: MLU_Morphemes ~ source_is_synthetic * clinical_group + factor(round_id) + (1 | cluster_id)
## m1_mlu: MLU_Morphemes ~ source_is_synthetic * clinical_group + Age_c + factor(round_id) + (1 | cluste
## m2_mlu: MLU_Morphemes ~ source_is_synthetic * clinical_group + Age_c + source_is_synthetic:Age_c + cl
## m3_mlu: MLU_Morphemes ~ source_is_synthetic * clinical_group * Age_c + factor(round_id) + (1 | cluste
##        npar    AIC    BIC  logLik -2*log(L)   Chisq Df Pr(>Chisq)
## m0_mlu    9 869.11 905.34 -425.56    851.11
## m1_mlu   10 860.10 900.36 -420.05    840.10 11.0123  1  0.0009051 ***
## m2_mlu   12 861.53 909.85 -418.77    837.53  2.5633  2  0.2775724
## m3_mlu   13 863.50 915.84 -418.75    837.50  0.0308  1  0.8607219
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
summary(m1_mlu)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula: MLU_Morphemes ~ source_is_synthetic * clinical_group + Age_c +
##     factor(round_id) + (1 | cluster_id)
##    Data: df_4pop
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##     860.1     900.4    -420.0     840.1       404
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.0217 -0.2982  0.0191  0.2454  5.1759
##
## Random effects:
##  Groups     Name        Variance Std.Dev.
##  cluster_id (Intercept) 0.5814   0.7625
##  Residual               0.3099   0.5566
## Number of obs: 414, groups:  cluster_id, 65
##
## Fixed effects:
##                                          Estimate Std. Error         df t value
## (Intercept)                              6.257557   0.142608 108.914108  43.879
## source_is_synthetic                      0.960308   0.266892  57.780723   3.598
## clinical_groupSLI                       -1.755378   0.278797  68.913400  -6.296
## Age_c                                    0.131094   0.038460  65.167046   3.409
## factor(round_id)round_02                 0.003059   0.113799 375.451900   0.027
## factor(round_id)round_03                -0.002636   0.113814 375.223508  -0.023
## factor(round_id)round_04                 0.009808   0.112668 370.037235   0.087
## source_is_synthetic:clinical_groupSLI    0.339635   0.288716  79.136547   1.176
##                                         Pr(>|t|)
## (Intercept)                              < 2e-16 ***
## source_is_synthetic                     0.000666 ***
## clinical_groupSLI                       2.45e-08 ***
## Age_c                                   0.001124 **
## factor(round_id)round_02                0.978572
## factor(round_id)round_03                0.981533
## factor(round_id)round_04                0.930677
## source_is_synthetic:clinical_groupSLI 0.242975
```

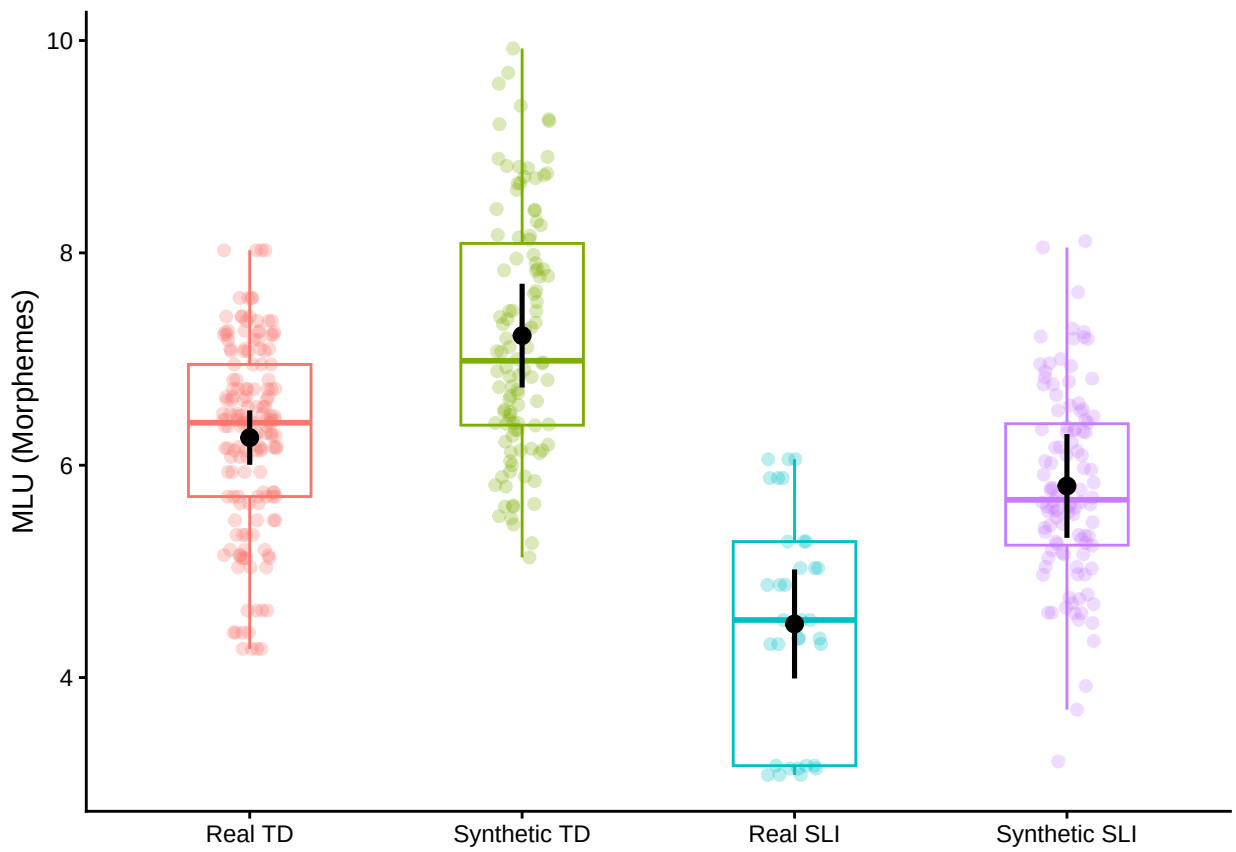
```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) src_s_ cl_SLI Age_c  f(_)_02 f(_)_03 f(_)_04
## src_s_synth -0.411
## clncl_grSLI -0.388  0.208
## Age_c       -0.008  0.023 -0.008
## fctr(r_)_02 -0.399  0.000  0.000 -0.011
## fctr(r_)_03 -0.399  0.000  0.000 -0.020  0.500
## fctr(r_)_04 -0.402  0.020 -0.004 -0.024  0.505   0.505
## src_s_:_SLI  0.375 -0.237 -0.966  0.003  0.000   0.000   0.004
```

Synthetic transcript have higher mlu than real ones. SLI lower than TD. Magnitude of difference is the same between synthetic SLI and TD. mlu increases with age.

- Intercept = predicted mlu for real TD in round 1 at mean age (centered)
- source_is_synthetic = TD synthetic data are about 0.96 morphemes higher
- clinical_groupSLI = real SLI is 1.76 morphemes lower
- Age_c = mlu increases with age
- source_is_synthetic:clinical_groupSLI = the difference between synthetic TD and SLI and real TD and SLI is not significant.

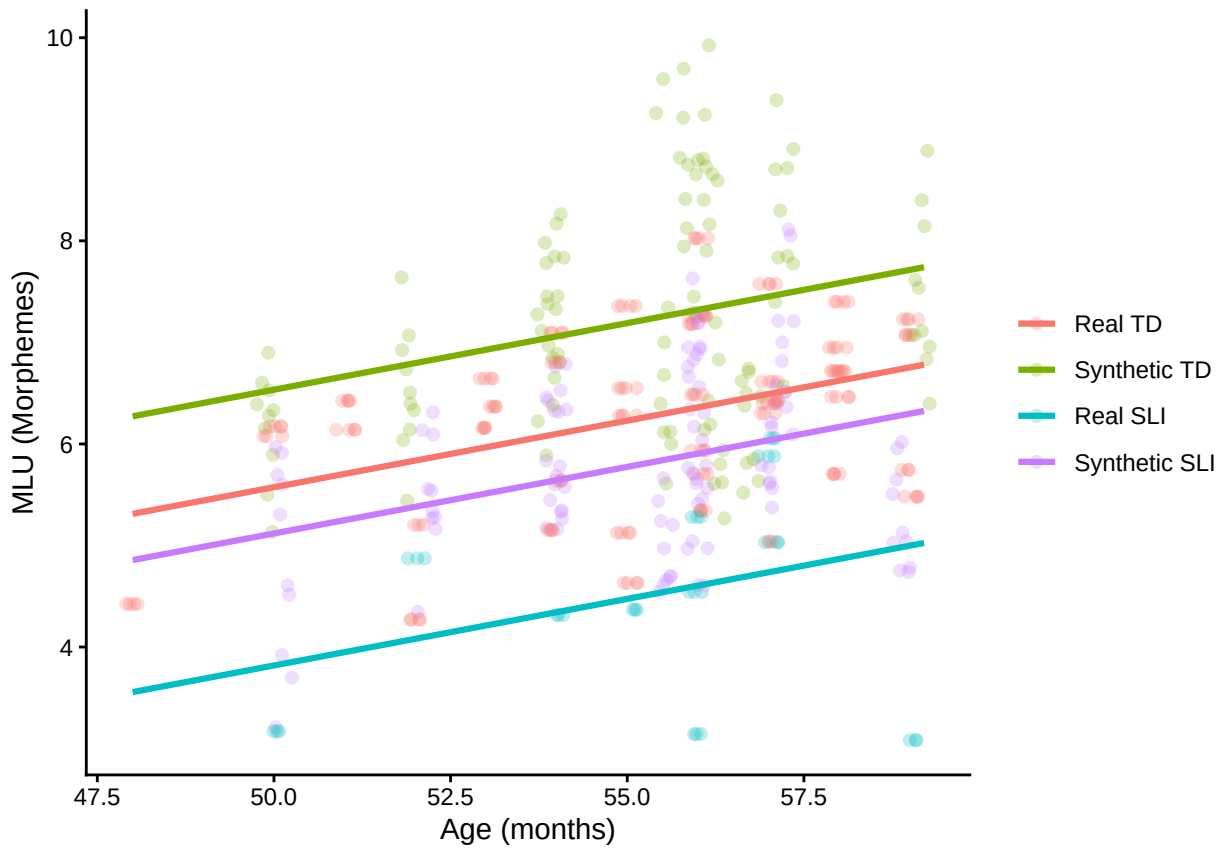
```r
library(dplyr)
library(ggplot2)
library(emmeans)

# If not already present
df_plot <- df_4pop %>%
  mutate(
    population = factor(
      paste(ifelse(source_is_synthetic == 1, "Synthetic", "Real"), clinical_group),
      levels = c("Real TD", "Synthetic TD", "Real SLI", "Synthetic SLI")
    )
  ) %>%
  filter(!is.na(MLU_Morphemes), !is.na(population))

# Estimated marginal means from the model
emm <- emmeans(m1_mlu, ~ source_is_synthetic * clinical_group)
emm_df <- as.data.frame(emm) %>%
  mutate(
    population = factor(
      paste(ifelse(source_is_synthetic == 1, "Synthetic", "Real"), clinical_group),
      levels = c("Real TD", "Synthetic TD", "Real SLI", "Synthetic SLI")
    )
  )

ggplot(df_plot, aes(x = population, y = MLU_Morphemes, color = population)) +
  geom_boxplot(width = 0.45, alpha = 0.15, outlier.shape = NA) +
  geom_jitter(width = 0.10, alpha = 0.28, size = 1.8) +
  geom_pointrange(
    data = emm_df,
    aes(x = population, y = emmean, ymin = lower.CL, ymax = upper.CL),
```

```r
    inherit.aes = FALSE,
    linewidth = 0.9
  ) +
  labs(
    x = NULL,
    y = "MLU (Morphemes)"
  ) +
  theme_classic() +
  theme(legend.position = "none")
```




```r
library(dplyr)
library(ggplot2)
library(emmeans)

age_mean <- mean(df_4pop$Age_Month, na.rm = TRUE)

df_plot <- df_4pop %>%
  mutate(
    Age_Month_plot = Age_c + age_mean,
    population = factor(
      paste(ifelse(source_is_synthetic == 1, "Synthetic", "Real"), clinical_group),
      levels = c("Real TD", "Synthetic TD", "Real SLI", "Synthetic SLI")
    )
  ) %>%
  filter(!is.na(MLU_Morphemes), !is.na(Age_c), !is.na(Age_Month_plot))
```

```r
age_grid <- seq(min(df_plot$Age_c), max(df_plot$Age_c), length.out = 100)

emm_age <- emmeans(
  m1_mlu,
  ~ source_is_synthetic * clinical_group * Age_c,
  at = list(Age_c = age_grid)
)

pred_df <- as.data.frame(emm_age) %>%
  mutate(
    Age_Month_plot = Age_c + age_mean,
    population = factor(
      paste(ifelse(source_is_synthetic == 1, "Synthetic", "Real"), clinical_group),
      levels = c("Real TD", "Synthetic TD", "Real SLI", "Synthetic SLI")
    )
  )

ggplot(df_plot, aes(x = Age_Month_plot, y = MLU_Morphemes, color = population)) +
  geom_jitter(width = 0.15, height = 0, alpha = 0.25, size = 1.8) +
  geom_line(
    data = pred_df,
    aes(y = emmean),
    linewidth = 1.1
  ) +
  labs(
    x = "Age (months)",
    y = "MLU (Morphemes)",
    color = NULL
  ) +
  theme_classic()
```



lexical diversity

calculates the diversity of a text by modelling how the Type-Token Ratio (TTR) changes as sample sizes increase.

```
m0_vocd <- lmer(
  VOCD_D_optimum_average ~ source_is_synthetic * clinical_group + factor(round_id) + (1 | cluster_id),
  data = df_4pop, REML = FALSE
)

m1_vocd <- lmer(
  VOCD_D_optimum_average ~ source_is_synthetic * clinical_group + Age_c + factor(round_id) + (1 | cluste
  data = df_4pop, REML = FALSE
)

m2_vocd <- lmer(
  VOCD_D_optimum_average ~ source_is_synthetic * clinical_group + Age_c +
    source_is_synthetic:Age_c + clinical_group:Age_c +
    factor(round_id) + (1 | cluster_id),
  data = df_4pop, REML = FALSE
) # exploratory model: do real vs syn and td vs sli differ? does the difference depend from clinical gr

m3_vocd <- lmer(
  VOCD_D_optimum_average ~ source_is_synthetic * clinical_group * Age_c +
    factor(round_id) + (1 | cluster_id),
```

```r
  data = df_4pop, REML = FALSE
)

anova(m0_vocd, m1_vocd, m2_vocd, m3_vocd)
```

```
## Data: df_4pop
## Models:
## m0_vocd: VOCD_D_optimum_average ~ source_is_synthetic * clinical_group + factor(round_id) + (1 | clus
## m1_vocd: VOCD_D_optimum_average ~ source_is_synthetic * clinical_group + Age_c + factor(round_id) +
## m2_vocd: VOCD_D_optimum_average ~ source_is_synthetic * clinical_group + Age_c + source_is_synthetic
## m3_vocd: VOCD_D_optimum_average ~ source_is_synthetic * clinical_group * Age_c + factor(round_id) +
##         npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
## m0_vocd    9 2836.5 2872.8 -1409.3    2818.5
## m1_vocd   10 2834.2 2874.4 -1407.1    2814.2 4.3212  1    0.03764 *
## m2_vocd   12 2833.1 2881.4 -1404.5    2809.1 5.1305  2    0.07690 .
## m3_vocd   13 2833.4 2885.8 -1403.7    2807.4 1.6104  1    0.20444
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
summary(m1_vocd)  # main model - do real vs syn and td vs sli differ? does the difference depend from c
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula: VOCD_D_optimum_average ~ source_is_synthetic * clinical_group +
##     Age_c + factor(round_id) + (1 | cluster_id)
##    Data: df_4pop
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    2834.2    2874.5   -1407.1    2814.2       404
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.2349 -0.3761 -0.0454  0.2528  4.3149
##
## Random effects:
##  Groups     Name        Variance Std.Dev.
##  cluster_id (Intercept) 41.97    6.478
##  Residual               39.35    6.273
## Number of obs: 414, groups:  cluster_id, 65
##
## Fixed effects:
##                                          Estimate Std. Error         df t value
## (Intercept)                             28.802851   1.350976 138.055101  21.320
## source_is_synthetic                     14.724513   2.324877  56.329761   6.333
## clinical_groupSLI                       -8.879408   2.496364  74.909947  -3.557
## Age_c                                    0.731449   0.341019  66.830872   2.145
## factor(round_id)round_02                 0.074829   1.265431 393.093060   0.059
## factor(round_id)round_03                -0.628045   1.265778 392.659699  -0.496
## factor(round_id)round_04                 0.008978   1.256414 385.329338   0.007
## source_is_synthetic:clinical_groupSLI    9.346252   2.635704  92.486050   3.546
##                                       Pr(>|t|)
## (Intercept)                            < 2e-16 ***
## source_is_synthetic                   4.24e-08 ***
```

```
## clinical_groupSLI                     0.000655 ***
## Age_c                                 0.035600 *
## factor(round_id)round_02              0.952876
## factor(round_id)round_03              0.620050
## factor(round_id)round_04              0.994303
## source_is_synthetic:clinical_groupSLI 0.000616 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) src_s_ cl_SLI Age_c  f(_)_02 f(_)_03 f(_)_04
## src_s_synth -0.396
## clncl_grSLI -0.361  0.210
## Age_c        0.002  0.023 -0.008
## fctr(r_)_02 -0.468  0.000  0.000 -0.018
## fctr(r_)_03 -0.468 -0.001  0.000 -0.029  0.500
## fctr(r_)_04 -0.471  0.025 -0.005 -0.036  0.504   0.504
## src_s_:_SLI  0.342 -0.258 -0.947  0.002  0.000   0.000   0.005
```

In the real data, SLI has lower lexical diversity than TD. In the synthetic data, lexical diversity is inflated for both groups, but especially for SLI, so the real TD–SLI difference is greatly reduced or effectively erased.

- Intercept = predicted lexical diversity for real TD in round 1 at mean age (centered)
- source_is_synthetic = TD synthetic data have a much higher lexical diversity than real transcripts.
- clinical_groupSLI = real SLI is 8.88 points lower than TD
- Age_c = lexical diversity increases with age
- source_is_synthetic:clinical_groupSLI = the synthetic-real difference is even larger in SLI than in TD.

```r
library(emmeans)

# Estimated marginal means at mean age, averaged over round
emm_vocd <- emmeans(
  m1_vocd,
  ~ source_is_synthetic * clinical_group,
  at = list(source_is_synthetic = c(0, 1))
)

emm_vocd
```

```
##  source_is_synthetic clinical_group emmean   SE   df lower.CL upper.CL
##                    0 TD               28.7 1.14 72.5     26.4     30.9
##                    1 TD               43.4 2.12 57.7     39.1     47.6
##                    0 SLI              19.8 2.31 78.8     15.2     24.4
##                    1 SLI              43.9 2.12 57.7     39.6     48.1
##
## Results are averaged over the levels of: round_id
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```

```r
pairs(
  emmeans(
    m1_vocd,
```

```r
    ~ source_is_synthetic | clinical_group,
    at = list(source_is_synthetic = c(0, 1))
  ),
  infer = c(TRUE, TRUE),
  adjust = "none"
)
```

```
## clinical_group = TD:
##  contrast                                    estimate   SE   df lower.CL
##  source_is_synthetic0 - source_is_synthetic1    -14.7 2.41 60.7    -19.5
##  upper.CL t.ratio p.value
##     -9.91  -6.112  <.0001
##
## clinical_group = SLI:
##  contrast                                    estimate   SE   df lower.CL
##  source_is_synthetic0 - source_is_synthetic1    -24.1 3.14 68.1    -30.3
##  upper.CL t.ratio p.value
##    -17.81  -7.676  <.0001
##
## Results are averaged over the levels of: round_id
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```

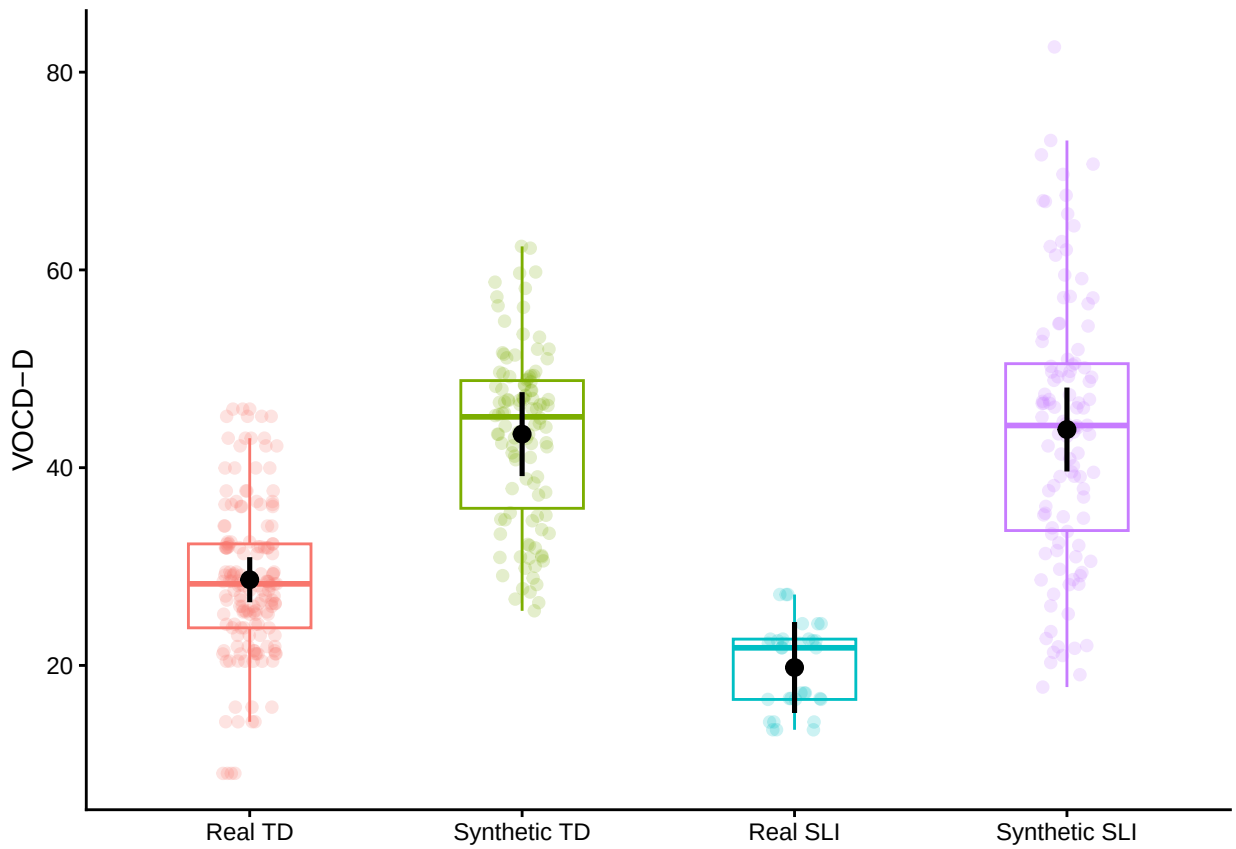
- real vs synthetic TD extremely different.
- real vs synthetic SLI extremely different.

```r
pairs(
  emmeans(
    m1_vocd,
    ~ clinical_group | source_is_synthetic,
    at = list(source_is_synthetic = c(0, 1))
  ),
  infer = c(TRUE, TRUE),
  adjust = "none"
)
```

```
## source_is_synthetic = 0:
##  contrast estimate    SE    df lower.CL upper.CL t.ratio p.value
##  TD - SLI    8.879 2.580  77.5     3.75    14.01   3.446  0.0009
##
## source_is_synthetic = 1:
##  contrast estimate    SE    df lower.CL upper.CL t.ratio p.value
##  TD - SLI   -0.467 0.851 353.1    -2.14     1.21  -0.549  0.5835
##
## Results are averaged over the levels of: round_id
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```

- significant difference between real TD and SLI.
- no difference between synthetic TD and SLI.

```r
library(dplyr)
library(ggplot2)
library(emmeans)

emm_vocd_df <- as.data.frame(emm_vocd) %>%
  mutate(
    population = factor(
      paste(ifelse(source_is_synthetic == 1, "Synthetic", "Real"), clinical_group),
      levels = c("Real TD", "Synthetic TD", "Real SLI", "Synthetic SLI")
    )
  )

df_plot <- df_4pop %>%
  mutate(
    population = factor(
      paste(ifelse(source_is_synthetic == 1, "Synthetic", "Real"), clinical_group),
      levels = c("Real TD", "Synthetic TD", "Real SLI", "Synthetic SLI")
    )
  ) %>%
  filter(!is.na(VOCD_D_optimum_average))

ggplot(df_plot, aes(x = population, y = VOCD_D_optimum_average, color = population)) +
  geom_boxplot(width = 0.45, alpha = 0.15, outlier.shape = NA) +
  geom_jitter(width = 0.10, alpha = 0.20, size = 1.7) +
  geom_pointrange(
    data = emm_vocd_df,
    aes(y = emmean, ymin = lower.CL, ymax = upper.CL),
    inherit.aes = TRUE,
    linewidth = 0.9,
    color = "black"
  ) +
  labs(
    x = NULL,
    y = "VOCD-D"
  ) +
  theme_classic() +
  theme(legend.position = "none")
```




```
df_4pop %>%
  mutate(
    population = paste(ifelse(source_is_synthetic == 1, "Synthetic", "Real"), clinical_group)
  ) %>%
  group_by(population) %>%
  summarise(
    raw_mean = mean(VOCD_D_optimum_average, na.rm = TRUE),
    raw_sd = sd(VOCD_D_optimum_average, na.rm = TRUE),
    n = n(),
    .groups = "drop"
  )
```

```
## # A tibble: 4 x 4
##   population    raw_mean raw_sd     n
##   <chr>            <dbl>  <dbl> <int>
## 1 Real SLI          19.9   4.29    33
## 2 Real TD           28.6   7.81   161
## 3 Synthetic SLI     43.8  13.6    110
## 4 Synthetic TD      43.3   8.74   110
```

Real SLI much lower than TD. Synthetic no difference.

```
summary(m2_vocd)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
```

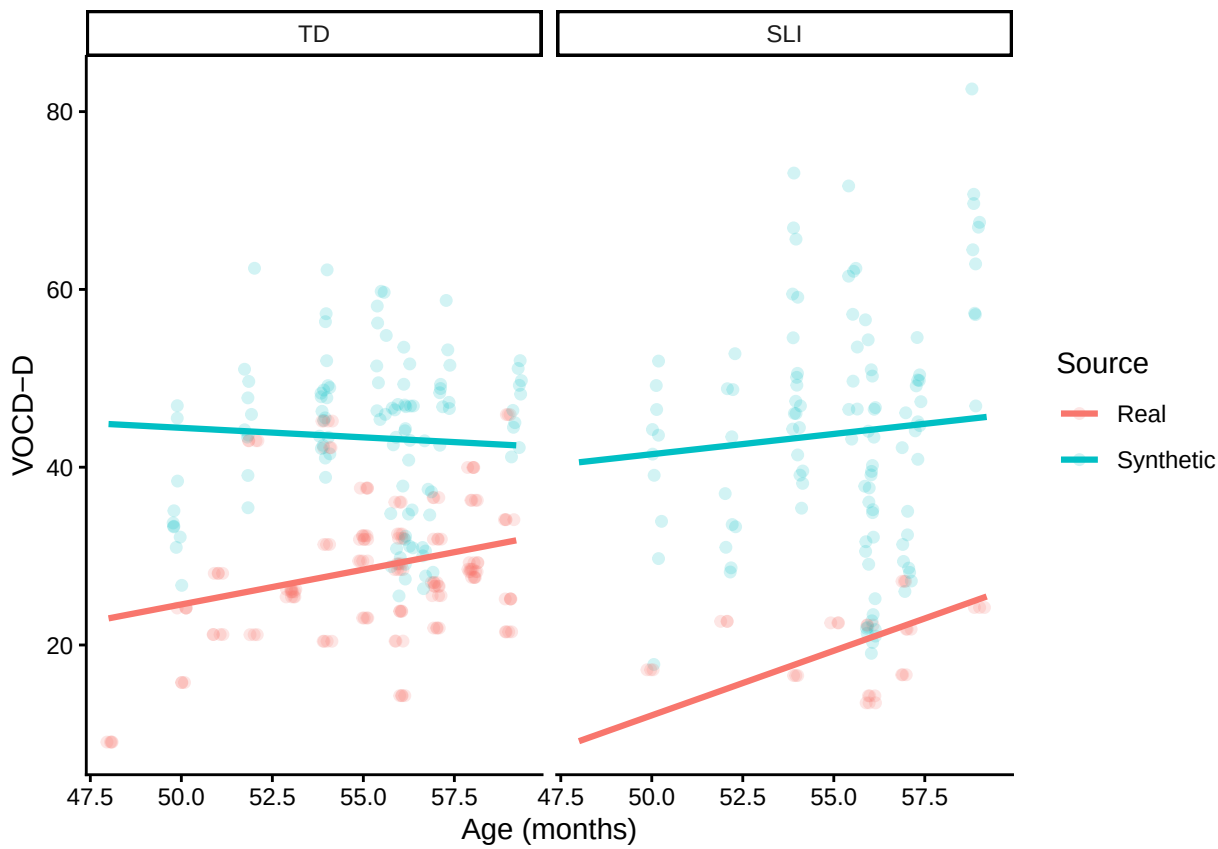
```
##   method [lmerModLmerTest]
## Formula: VOCD_D_optimum_average ~ source_is_synthetic * clinical_group +
##     Age_c + source_is_synthetic:Age_c + clinical_group:Age_c +
##     factor(round_id) + (1 | cluster_id)
##    Data: df_4pop
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    2833.1    2881.4   -1404.5    2809.1       402
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.9996 -0.3582 -0.0428  0.2347  4.1424
##
## Random effects:
##  Groups     Name        Variance Std.Dev.
##  cluster_id (Intercept) 43.43    6.590
##  Residual               38.60    6.213
## Number of obs: 414, groups:  cluster_id, 65
##
## Fixed effects:
##                                      Estimate Std. Error       df t value
## (Intercept)                           28.6571     1.3667 140.2682  20.968
## source_is_synthetic                   14.6444     2.3587  55.5501   6.209
## clinical_groupSLI                     -8.9722     2.5245  73.0899  -3.554
## Age_c                                  0.7837     0.3840  71.2671   2.041
## factor(round_id)round_02               0.2146     1.2633 383.2724   0.170
## factor(round_id)round_03              -0.4127     1.2701 376.3609  -0.325
## factor(round_id)round_04               0.2257     1.2662 361.0691   0.178
## source_is_synthetic:clinical_groupSLI   9.5134     2.6605  89.6667   3.576
## source_is_synthetic:Age_c             -0.9975     0.9266  58.9375  -1.076
## clinical_groupSLI:Age_c                0.6697     0.3314 408.6225   2.021
##                                      Pr(>|t|)
## (Intercept)                           < 2e-16 ***
## source_is_synthetic                  7.14e-08 ***
## clinical_groupSLI                    0.000669 ***
## Age_c                                0.044977 *
## factor(round_id)round_02             0.865209
## factor(round_id)round_03             0.745429
## factor(round_id)round_04             0.858621
## source_is_synthetic:clinical_groupSLI 0.000565 ***
## source_is_synthetic:Age_c            0.286093
## clinical_groupSLI:Age_c              0.043932 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) src_s_ cl_SLI Age_c  f(_)_02 f(_)_03 f(_)_04 s__:_S s__:A_
## src_s_synth -0.392
## clncl_grSLI -0.361  0.210
## Age_c       -0.046  0.012 -0.007
## fctr(r_)_02 -0.469 -0.003  0.000  0.032
## fctr(r_)_03 -0.471 -0.004  0.000  0.038  0.508
## fctr(r_)_04 -0.475  0.021 -0.006  0.045  0.513   0.518
## src_s_:_SLI  0.342 -0.256 -0.949  0.005  0.002   0.002   0.008
```

```
## src_s_sy:A_  0.111  0.023  0.005 -0.413 -0.113  -0.151  -0.178  -0.016
## clnc_SLI:A_  0.007 -0.004 -0.016 -0.143 -0.003  -0.007  -0.009   0.023 -0.006
```

- vocd-d rises with age for real TD and even more for SLI
- flat for synthetic TD and rises a little bit for SLI

```r
library(dplyr)
library(ggplot2)
library(emmeans)

# Mean age used to convert centered age back to months for plotting
age_mean <- mean(df_4pop$Age_Month, na.rm = TRUE)

# Raw data for plotting
df_plot <- df_4pop %>%
  mutate(
    Age_Month_plot = Age_c + age_mean,
    source_label = factor(
      source_is_synthetic,
      levels = c(0, 1),
      labels = c("Real", "Synthetic")
    ),
    clinical_group = factor(clinical_group, levels = c("TD", "SLI"))
  ) %>%
  filter(
    !is.na(VOCD_D_optimum_average),
    !is.na(Age_c),
    !is.na(Age_Month_plot)
  )

# Age grid for predictions
age_grid <- seq(
  min(df_plot$Age_c, na.rm = TRUE),
  max(df_plot$Age_c, na.rm = TRUE),
  length.out = 100
)

# Model-based predictions from m2_vocd
emm_age <- emmeans(
  m2_vocd,
  ~ source_is_synthetic * clinical_group * Age_c,
  at = list(
    source_is_synthetic = c(0, 1),
    Age_c = age_grid
  )
)

pred_df <- as.data.frame(emm_age) %>%
  mutate(
    Age_Month_plot = Age_c + age_mean,
    source_label = factor(
      source_is_synthetic,
      levels = c(0, 1),
      labels = c("Real", "Synthetic")
```

```r
    ),
    clinical_group = factor(clinical_group, levels = c("TD", "SLI"))
  )

# Plot
ggplot(df_plot, aes(x = Age_Month_plot, y = VOCD_D_optimum_average, color = source_label)) +
  geom_jitter(width = 0.15, height = 0, alpha = 0.18, size = 1.6) +
  geom_line(
    data = pred_df,
    aes(x = Age_Month_plot, y = emmean, color = source_label),
    linewidth = 1.1,
    inherit.aes = FALSE
  ) +
  facet_wrap(~ clinical_group) +
  labs(
    x = "Age (months)",
    y = "VOCD-D",
    color = "Source"
  ) +
  theme_classic()
```



source_is_synthetic:age is not significant, so the downward slope of TD is to interpret cautiously.

developmental sentence score

```r
m0_dss <- lmer(
  DSS_tbl_score ~ source_is_synthetic * clinical_group + factor(round_id) + (1 | cluster_id),
  data = df_4pop, REML = FALSE
)

m1_dss <- lmer(
  DSS_tbl_score ~ source_is_synthetic * clinical_group + Age_c + factor(round_id) + (1 | cluster_id),
  data = df_4pop, REML = FALSE
)

m2_dss <- lmer(
  DSS_tbl_score ~ source_is_synthetic * clinical_group + Age_c +
    source_is_synthetic:Age_c + clinical_group:Age_c +
    factor(round_id) + (1 | cluster_id),
  data = df_4pop, REML = FALSE
)

m3_dss <- lmer(
  DSS_tbl_score ~ source_is_synthetic * clinical_group * Age_c + factor(round_id) + (1 | cluster_id),
  data = df_4pop, REML = FALSE
)

anova(m0_dss, m1_dss, m2_dss, m3_dss)
```

```
## Data: df_4pop
## Models:
## m0_dss: DSS_tbl_score ~ source_is_synthetic * clinical_group + factor(round_id) + (1 | cluster_id)
## m1_dss: DSS_tbl_score ~ source_is_synthetic * clinical_group + Age_c + factor(round_id) + (1 | cluste
## m2_dss: DSS_tbl_score ~ source_is_synthetic * clinical_group + Age_c + source_is_synthetic:Age_c + cl
## m3_dss: DSS_tbl_score ~ source_is_synthetic * clinical_group * Age_c + factor(round_id) + (1 | cluste
##        npar    AIC    BIC  logLik -2*log(L)   Chisq Df Pr(>Chisq)
## m0_dss    9 1182.2 1218.5 -582.11    1164.2
## m1_dss   10 1172.3 1212.6 -576.18    1152.3 11.8718  1  0.0005699 ***
## m2_dss   12 1171.7 1220.0 -573.87    1147.7  4.6225  2  0.0991359 .
## m3_dss   13 1173.7 1226.1 -573.87    1147.7  0.0002  1  0.9887766
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
summary(m1_dss)
```

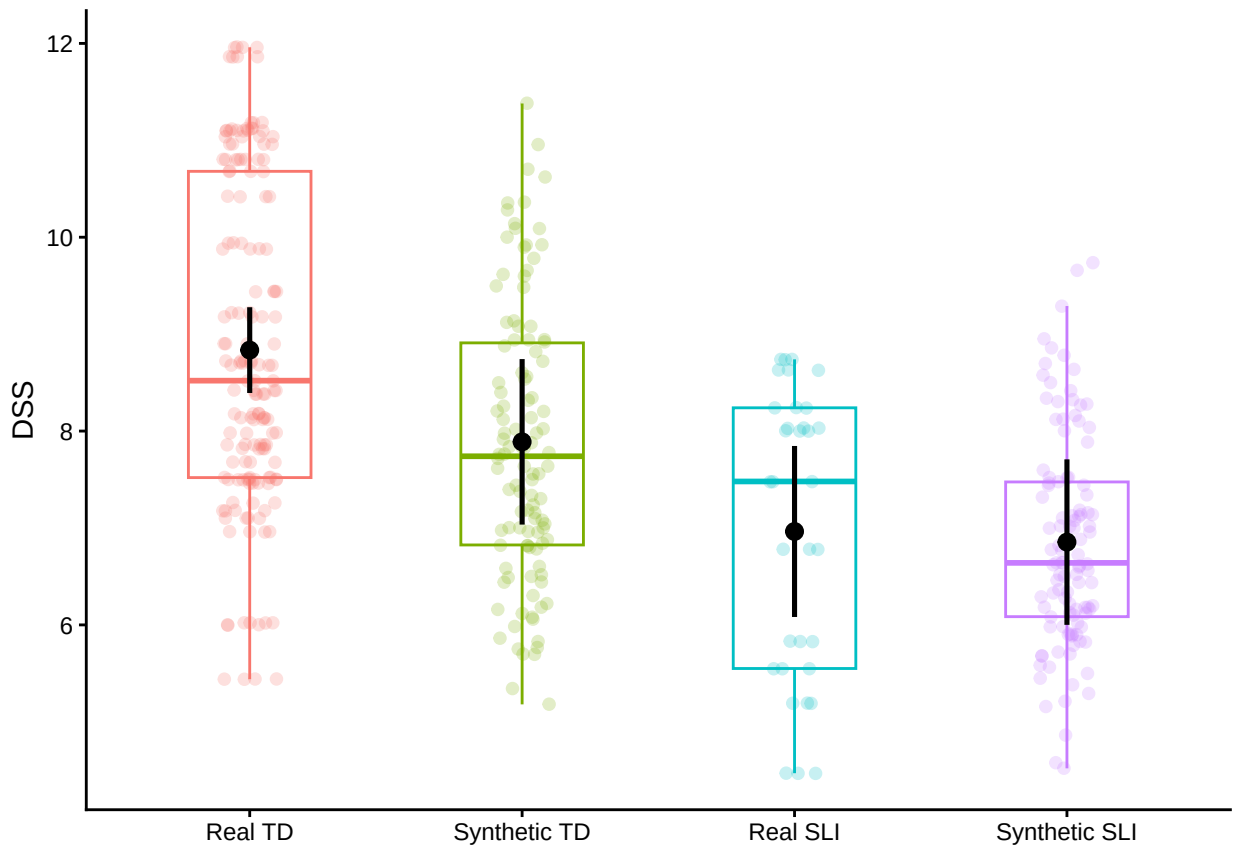
```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula: DSS_tbl_score ~ source_is_synthetic * clinical_group + Age_c +
##     factor(round_id) + (1 | cluster_id)
##    Data: df_4pop
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    1172.4    1212.6    -576.2    1152.4       404
##
## Scaled residuals:
```

```
##     Min      1Q  Median      3Q     Max
## -2.6583 -0.3076 -0.0254  0.2383  3.8223
##
## Random effects:
##  Groups     Name        Variance Std.Dev.
##  cluster_id (Intercept) 1.8136   1.3467
##  Residual               0.6181   0.7862
## Number of obs: 414, groups:  cluster_id, 65
##
## Fixed effects:
##                                          Estimate Std. Error         df t value
## (Intercept)                             8.842e+00  2.365e-01  9.348e+01  37.393
## source_is_synthetic                    -9.466e-01  4.656e-01  5.933e+01  -2.033
## clinical_groupSLI                      -1.871e+00  4.793e-01  6.672e+01  -3.905
## Age_c                                   2.348e-01  6.636e-02  6.538e+01   3.538
## factor(round_id)round_02               -1.034e-02  1.618e-01  3.658e+02  -0.064
## factor(round_id)round_03               -1.530e-02  1.618e-01  3.657e+02  -0.095
## factor(round_id)round_04                5.686e-05  1.600e-01  3.621e+02   0.000
## source_is_synthetic:clinical_groupSLI  8.369e-01  4.908e-01  7.337e+01   1.705
##                                        Pr(>|t|)
## (Intercept)                             < 2e-16 ***
## source_is_synthetic                    0.046524 *
## clinical_groupSLI                      0.000222 ***
## Age_c                                  0.000747 ***
## factor(round_id)round_02               0.949078
## factor(round_id)round_03               0.924757
## factor(round_id)round_04               0.999717
## source_is_synthetic:clinical_groupSLI 0.092401 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) src_s_ cl_SLI Age_c  f(_)_02 f(_)_03 f(_)_04
## src_s_synth -0.422
## clncl_grSLI -0.406  0.206
## Age_c       -0.012  0.023 -0.008
## fctr(r_)_02 -0.342  0.000  0.000 -0.008
## fctr(r_)_03 -0.342  0.000  0.000 -0.015  0.500
## fctr(r_)_04 -0.344  0.016 -0.004 -0.018  0.506   0.506
## src_s_:_SLI  0.396 -0.226 -0.976  0.003  0.000   0.000   0.004
```

Synthetic transcripts have lower DSS than real ones. SLI < TD. DSS increases with age. synthetic vs real gap smaller in SLI than TD (only marginally significant). The synthetic data do not erase the clinical-group difference; they preserve it fairly well.

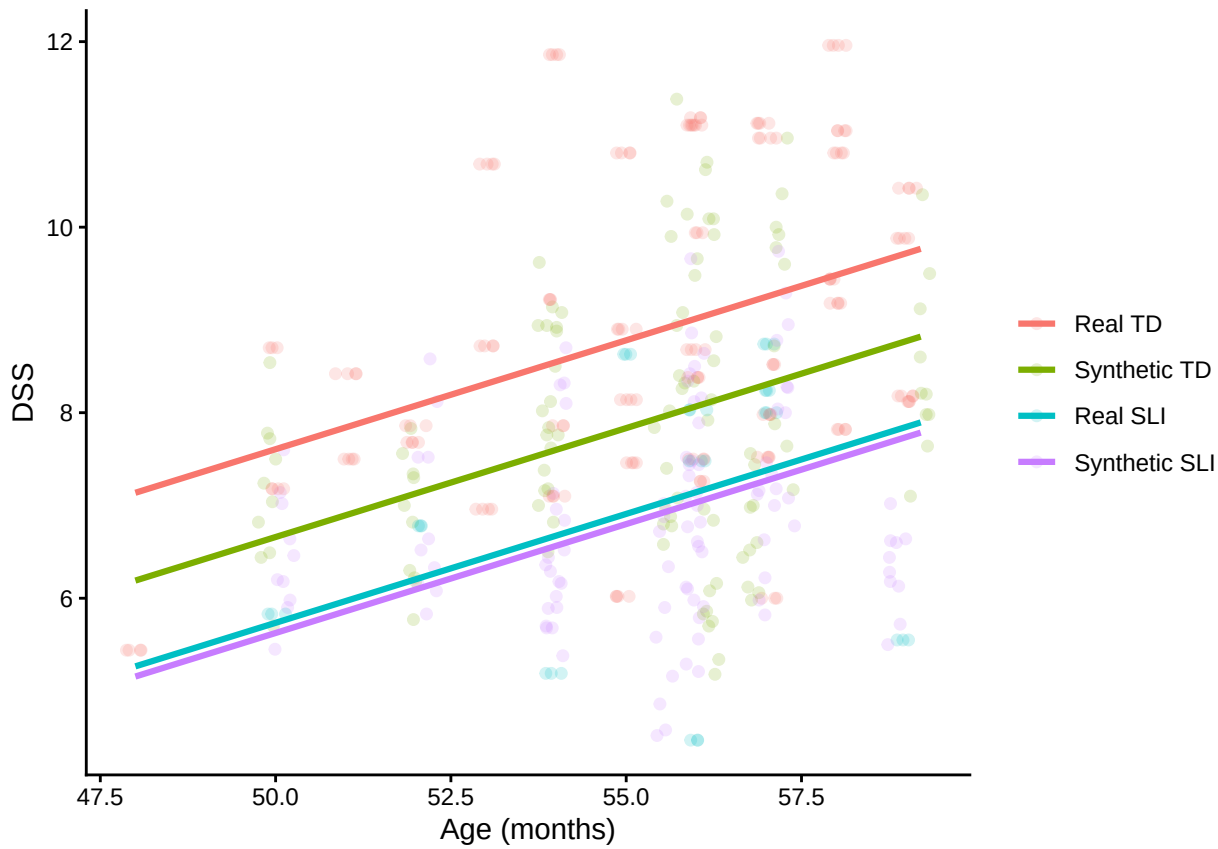
- Intercept = Predicted DSS for real TD in round 01 at mean age.
- source_is_synthetic = In the TD group, synthetic transcripts are about 0.95 DSS points lower than real transcripts.
- clinical_groupSLI = Among real transcripts, SLI is about 1.87 DSS points lower than TD.
- Age_c = DSS increases with age.
- source_is_synthetic:clinical_groupSLI = This suggests the synthetic penalty may be less negative in SLI than in TD, but the evidence is weak.

```r
# Adjusted means from the retained model
emm_dss <- emmeans(
  m1_dss,
  ~ source_is_synthetic * clinical_group,
  at = list(source_is_synthetic = c(0, 1))
)

emm_dss_df <- as.data.frame(emm_dss) %>%
  mutate(
    population = factor(
      paste(ifelse(source_is_synthetic == 1, "Synthetic", "Real"), clinical_group),
      levels = c("Real TD", "Synthetic TD", "Real SLI", "Synthetic SLI")
    )
  )

# Raw data for plotting
df_plot <- df_4pop %>%
  mutate(
    population = factor(
      paste(ifelse(source_is_synthetic == 1, "Synthetic", "Real"), clinical_group),
      levels = c("Real TD", "Synthetic TD", "Real SLI", "Synthetic SLI")
    )
  ) %>%
  filter(!is.na(DSS_tbl_score), !is.na(population))

# Plot
ggplot(df_plot, aes(x = population, y = DSS_tbl_score, color = population)) +
  geom_boxplot(width = 0.45, alpha = 0.15, outlier.shape = NA) +
  geom_jitter(width = 0.10, alpha = 0.22, size = 1.7) +
  geom_pointrange(
    data = emm_dss_df,
    aes(x = population, y = emmean, ymin = lower.CL, ymax = upper.CL),
    inherit.aes = FALSE,
    linewidth = 0.9,
    color = "black"
  ) +
  labs(
    x = NULL,
    y = "DSS"
  ) +
  theme_classic() +
  theme(legend.position = "none")
```




```
library(dplyr)
library(ggplot2)
library(emmeans)

age_mean <- mean(df_4pop$Age_Month, na.rm = TRUE)

df_age_plot <- df_4pop %>%
  mutate(
    Age_Month_plot = Age_c + age_mean,
    population = factor(
      paste(ifelse(source_is_synthetic == 1, "Synthetic", "Real"), clinical_group),
      levels = c("Real TD", "Synthetic TD", "Real SLI", "Synthetic SLI")
    )
  ) %>%
  filter(!is.na(DSS_tbl_score), !is.na(Age_c), !is.na(Age_Month_plot))

age_grid <- seq(
  min(df_age_plot$Age_c, na.rm = TRUE),
  max(df_age_plot$Age_c, na.rm = TRUE),
  length.out = 100
)

emm_age <- emmeans(
  m1_dss,
  ~ source_is_synthetic * clinical_group * Age_c,
  at = list(
```

```r
    source_is_synthetic = c(0, 1),
    Age_c = age_grid
  )
)

pred_df <- as.data.frame(emm_age) %>%
  mutate(
    Age_Month_plot = Age_c + age_mean,
    population = factor(
      paste(ifelse(source_is_synthetic == 1, "Synthetic", "Real"), clinical_group),
      levels = c("Real TD", "Synthetic TD", "Real SLI", "Synthetic SLI")
    )
  )

ggplot(df_age_plot, aes(x = Age_Month_plot, y = DSS_tbl_score, color = population)) +
  geom_jitter(width = 0.15, height = 0, alpha = 0.18, size = 1.6) +
  geom_line(
    data = pred_df,
    aes(x = Age_Month_plot, y = emmean, color = population),
    linewidth = 1.1
  ) +
  labs(
    x = "Age (months)",
    y = "DSS",
    color = NULL
  ) +
  theme_classic()
```



* all groups dss increase with age * main difference is in the distance between real and synthetic TD, not different developmental slopes.

```
summary(m2_dss)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula: DSS_tbl_score ~ source_is_synthetic * clinical_group + Age_c +
##     source_is_synthetic:Age_c + clinical_group:Age_c + factor(round_id) +
##     (1 | cluster_id)
##    Data: df_4pop
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    1171.7    1220.0    -573.9    1147.7       402
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.7973 -0.2841 -0.0273  0.2271  3.8245
##
## Random effects:
##  Groups     Name        Variance Std.Dev.
##  cluster_id (Intercept) 1.8026   1.3426
##  Residual               0.6108   0.7815
## Number of obs: 414, groups:  cluster_id, 65
##
## Fixed effects:
```

```
##                                        Estimate Std. Error         df t value
## (Intercept)                             8.842e+00  2.359e-01  9.368e+01  37.473
## source_is_synthetic                    -9.439e-01  4.642e-01  5.883e+01  -2.033
## clinical_groupSLI                      -1.860e+00  4.777e-01  6.615e+01  -3.893
## Age_c                                   2.414e-01  7.288e-02  6.570e+01   3.313
## factor(round_id)round_02               -1.094e-02  1.613e-01  3.616e+02  -0.068
## factor(round_id)round_03               -1.545e-02  1.616e-01  3.585e+02  -0.096
## factor(round_id)round_04               -2.469e-04  1.601e-01  3.520e+02  -0.002
## source_is_synthetic:clinical_groupSLI  8.193e-01  4.893e-01  7.274e+01   1.675
## source_is_synthetic:Age_c               1.719e-02  1.794e-01  6.499e+01   0.096
## clinical_groupSLI:Age_c                -9.268e-02  4.344e-02  3.983e+02  -2.134
##                                        Pr(>|t|)
## (Intercept)                             < 2e-16 ***
## source_is_synthetic                    0.046561 *
## clinical_groupSLI                      0.000233 ***
## Age_c                                  0.001504 **
## factor(round_id)round_02               0.945955
## factor(round_id)round_03               0.923919
## factor(round_id)round_04               0.998770
## source_is_synthetic:clinical_groupSLI 0.098319 .
## source_is_synthetic:Age_c              0.923950
## clinical_groupSLI:Age_c                0.033490 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) src_s_ cl_SLI Age_c  f(_)_02 f(_)_03 f(_)_04 s__:_S s__:A_
## src_s_synth -0.420
## clncl_grSLI -0.405  0.206
## Age_c       -0.035  0.011 -0.008
## fctr(r_)_02 -0.344 -0.002  0.000  0.022
## fctr(r_)_03 -0.345 -0.003  0.000  0.027  0.503
## fctr(r_)_04 -0.347  0.013 -0.004  0.031  0.509   0.511
## src_s_:_SLI  0.395 -0.226 -0.976  0.008  0.001   0.001   0.005
## src_s_sy:A_  0.057  0.025  0.003 -0.416 -0.071  -0.095  -0.112  -0.014
## clnc_SLI:A_  0.008  0.001 -0.011 -0.101 -0.008  -0.013  -0.015   0.015  0.100
```

The model seems to indicate that there is some evidence that DSS increases more slowly with age in SLI than TD:

- TD shows a steeper age-related increase in DSS
- SLI shows a shallower increase
- real and synthetic are very similar in slope

Productive Morphology

past over-generalisation

Overgeneralization is primarily a developmental phenomenon tied to productive morphology. These errors appear after acquisition of regular past tense is completed. In SLI/DLD, the better-established clinical pattern is reduced tense marking and increased zero-marking, not necessarily increased overgeneralization. They lag behind past acquisition therefore, they do not yet showcase this error pattern.

Given a past-tense opportunity, what is the probability of producing an overgeneralized past form?

```r
library(dplyr)
library(tidyr)

# Read DEV data
real_dev <- read.csv(
  "/Users/shamiraventurini/PycharmProjects/ICL-PILOT/phase2/ENNI/measures/dev_measures.csv",
  check.names = FALSE
)

syn_dev <- read.csv(
  "/Users/shamiraventurini/PycharmProjects/ICL-PILOT/phase2/generated/measures/dev_measures.csv",
  check.names = FALSE
)

# Use existing synthetic metadata to recover round-specific prompt children.
# If `syn` is already in memory from your previous cell, use that.
# Otherwise fall back to syn_dev if it also contains the prompt columns.
prompt_source <- if (exists("syn")) syn else syn_dev

prompt_map <- prompt_source %>%
  transmute(
    round_id = as.character(round_id),
    prompt_td_child_id = as.character(prompt_td_child_id),
    prompt_sli_child_id = as.character(prompt_sli_child_id)
  ) %>%
  distinct() %>%
  pivot_longer(
    cols = c(prompt_td_child_id, prompt_sli_child_id),
    names_to = "prompt_group",
    values_to = "child_id"
  ) %>%
  filter(!is.na(child_id), child_id != "") %>%
  mutate(
    clinical_group = case_when(
      prompt_group == "prompt_td_child_id"  ~ "TD",
      prompt_group == "prompt_sli_child_id" ~ "SLI"
    )
  ) %>%
  select(round_id, clinical_group, child_id) %>%
  distinct()

# Real DEV pool
real_dev_4yo <- real_dev %>%
  mutate(
    File_ID = as.character(File_ID),
    Age_Month = as.numeric(`Age(Month)`),
    source = "real",
    pair_id = NA_character_,
    round_id = NA_character_,
    clinical_group = case_when(
      Group == "TD"  ~ "TD",
      Group == "SLI" ~ "SLI",
      TRUE ~ NA_character_
```

```r
    )
  ) %>%
  filter(
    !is.na(clinical_group),
    Age_Month >= 48, Age_Month < 60
  )

# Synthetic DEV pool
syn_dev_4yo <- syn_dev %>%
  mutate(
    pair_id = as.character(pair_id),
    round_id = as.character(round_id),
    Age_Month = as.numeric(`Age(Month)`),
    source = "synthetic",
    clinical_group = case_when(
      comparison_target_group == "TD"  ~ "TD",
      comparison_target_group == "SLI" ~ "SLI",
      TRUE ~ NA_character_
    )
  ) %>%
  filter(
    !is.na(clinical_group),
    Age_Month >= 48, Age_Month < 60
  )

# Build fold-stacked held-out DEV dataframe
rounds <- sort(unique(syn_dev_4yo$round_id))

df_dev_4pop <- bind_rows(lapply(rounds, function(r) {
  prompt_r <- prompt_map %>%
    filter(round_id == r)

  real_r <- real_dev_4yo %>%
    mutate(round_id = r) %>%
    anti_join(
      prompt_r,
      by = c("File_ID" = "child_id", "clinical_group")
    )

  syn_r <- syn_dev_4yo %>%
    filter(round_id == r)

  bind_rows(real_r, syn_r)
})) %>%
  mutate(
    source = factor(source, levels = c("real", "synthetic")),
    clinical_group = factor(clinical_group, levels = c("TD", "SLI")),
    source_is_synthetic = if_else(source == "synthetic", 1L, 0L),
    cluster_id = if_else(
      source == "synthetic",
      paste0("synthetic_", round_id, "_", pair_id),
      paste0("real_", File_ID)
    ),
```

```r
    Age_c = Age_Month - mean(Age_Month, na.rm = TRUE)
  ) %>%
  filter(
    !is.na(past_overgeneralization_errors),
    !is.na(past_verb_opportunities),
    past_verb_opportunities > 0
  )

# Quick checks
table(df_dev_4pop$source, df_dev_4pop$clinical_group)
```

```
##
##              TD SLI
##   real      161  33
##   synthetic 109 109
```

```r
summary(df_dev_4pop$past_overgeneralization_errors)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   0.000   0.000   0.000   1.262   2.000  12.000
```

```r
summary(df_dev_4pop$past_verb_opportunities)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    1.00   13.00   27.00   30.99   44.00  125.00
```

```r
m0_overgen <- glmer(
  cbind(past_overgeneralization_errors,
        past_verb_opportunities - past_overgeneralization_errors) ~
    source_is_synthetic * clinical_group + factor(round_id) + (1 | cluster_id),
  data = df_dev_4pop,
  family = binomial,
  control = glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 2e5))
) # Does the probability of past overgeneralization differ by source (real vs synthetic), by clinical g

m1_overgen <- glmer(
  cbind(past_overgeneralization_errors,
        past_verb_opportunities - past_overgeneralization_errors) ~
    source_is_synthetic * clinical_group + Age_c + factor(round_id) + (1 | cluster_id),
  data = df_dev_4pop,
  family = binomial,
  control = glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 2e5))
) # Does the probability of past overgeneralization change with age overall?

m2_overgen <- glmer(
  cbind(past_overgeneralization_errors,
        past_verb_opportunities - past_overgeneralization_errors) ~
    source_is_synthetic * clinical_group + Age_c +
    source_is_synthetic:Age_c + clinical_group:Age_c +
    factor(round_id) + (1 | cluster_id),
  data = df_dev_4pop,
```

```r
  family = binomial,
  control = glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 2e5))
) # Do real and synthetic data show different age slopes, or do TD and SLI show different age slopes?

m3_overgen <- glmer(
  cbind(past_overgeneralization_errors,
        past_verb_opportunities - past_overgeneralization_errors) ~
    source_is_synthetic * clinical_group * Age_c +
    factor(round_id) + (1 | cluster_id),
  data = df_dev_4pop,
  family = binomial,
  control = glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 2e5))
) # Is the developmental pattern of past overgeneralization across age and clinical group reproduced di

anova(m0_overgen, m1_overgen, m2_overgen, m3_overgen, test = "Chisq")
```

```
## Data: df_dev_4pop
## Models:
## m0_overgen: cbind(past_overgeneralization_errors, past_verb_opportunities - past_overgeneralization_
## m1_overgen: cbind(past_overgeneralization_errors, past_verb_opportunities - past_overgeneralization_
## m2_overgen: cbind(past_overgeneralization_errors, past_verb_opportunities - past_overgeneralization_
## m3_overgen: cbind(past_overgeneralization_errors, past_verb_opportunities - past_overgeneralization_
##            npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
## m0_overgen    8 767.99 800.16 -376.00    751.99
## m1_overgen    9 762.86 799.05 -372.43    744.86 7.1298  1   0.007581 **
## m2_overgen   11 766.19 810.42 -372.09    744.19 0.6709  2   0.715014
## m3_overgen   12 763.25 811.50 -369.62    739.25 4.9409  1   0.026229 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
summary(m3_overgen)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula: cbind(past_overgeneralization_errors, past_verb_opportunities -
##     past_overgeneralization_errors) ~ source_is_synthetic * clinical_group *
##     Age_c + factor(round_id) + (1 | cluster_id)
##    Data: df_dev_4pop
## Control: glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 2e+05))
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##     763.2     811.5    -369.6     739.2       400
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -0.8319 -0.3092 -0.1383  0.0386  9.0958
##
## Random effects:
##  Groups     Name        Variance Std.Dev.
##  cluster_id (Intercept) 1.44     1.2
## Number of obs: 412, groups:  cluster_id, 65
```

```
##
## Fixed effects:
##                                            Estimate Std. Error z value
## (Intercept)                               -2.920964   0.223269 -13.083
## source_is_synthetic                       -3.855720   0.627213  -6.147
## clinical_groupSLI                         -0.728614   0.544087  -1.339
## Age_c                                     -0.160135   0.073938  -2.166
## factor(round_id)round_02                  -0.001781   0.140707  -0.013
## factor(round_id)round_03                  -0.000788   0.138689  -0.006
## factor(round_id)round_04                  -0.003271   0.138892  -0.024
## source_is_synthetic:clinical_groupSLI      2.524771   0.733927   3.440
## source_is_synthetic:Age_c                 -0.156711   0.248162  -0.631
## clinical_groupSLI:Age_c                   -0.250987   0.205741  -1.220
## source_is_synthetic:clinical_groupSLI:Age_c  0.647389   0.301451   2.148
##                                           Pr(>|z|)
## (Intercept)                                < 2e-16 ***
## source_is_synthetic                       7.88e-10 ***
## clinical_groupSLI                         0.180522
## Age_c                                     0.030328 *
## factor(round_id)round_02                  0.989903
## factor(round_id)round_03                  0.995467
## factor(round_id)round_04                  0.981212
## source_is_synthetic:clinical_groupSLI     0.000582 ***
## source_is_synthetic:Age_c                 0.527722
## clinical_groupSLI:Age_c                   0.222497
## source_is_synthetic:clinical_groupSLI:Age_c 0.031747 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##            (Intr) src_s_ cl_SLI Age_c  f(_)_02 f(_)_03 f(_)_04 sr__:_SLI
## src_s_synth -0.277
## clncl_grSLI -0.300  0.125
## Age_c        0.009  0.002  0.006
## fctr(r_)_02 -0.320 -0.003 -0.007  0.011
## fctr(r_)_03 -0.320  0.006 -0.001  0.014  0.520
## fctr(r_)_04 -0.324  0.010 -0.005  0.015  0.520   0.529
## src_s_:_SLI  0.222 -0.505 -0.741 -0.004  0.010   0.003   0.005
## src_s_sy:A_  0.022  0.292 -0.004 -0.300 -0.052  -0.065  -0.074  -0.238
## clnc_SLI:A_  0.008  0.006  0.205 -0.353  0.019   0.023   0.021  -0.152
## s__:_SLI:A_ -0.009 -0.260 -0.142  0.240 -0.007  -0.016  -0.020   0.279
##            s__:A_ c_SLI:
## src_s_synth
## clncl_grSLI
## Age_c
## fctr(r_)_02
## fctr(r_)_03
## fctr(r_)_04
## src_s_:_SLI
## src_s_sy:A_
## clnc_SLI:A_  0.102
## s__:_SLI:A_ -0.465 -0.684
```

The predicted probabilities were low overall, but they were substantially lower in the synthetic data than in

the real data. That fits the literature too: overgeneralization is a real developmental phenomenon, but in spontaneous language samples it is usually infrequent, not something that happens constantly.

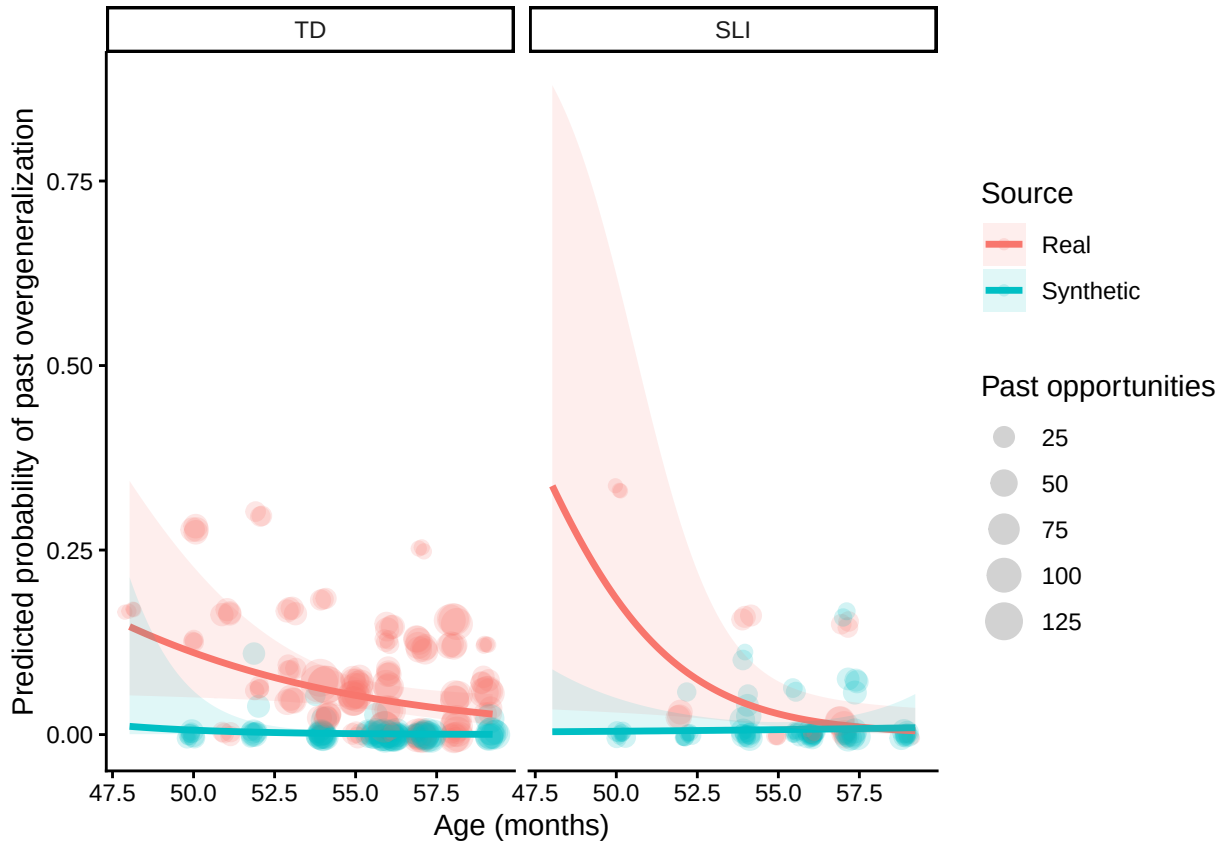
Past over-generalisation is much rarer in the synthetic data, especially TD. Real TD > SLI. At mean age, the model estimates lower past overgeneralization in real SLI than in real TD, but this baseline group difference is not statistically significant on its own, and the full model indicates that group differences vary with age and source. The relationship between overgeneralization, age, and clinical group is different in the synthetic data than in the real data. The main signal is about mismatch between real and synthetic developmental patterns, not a simple diagnostic-group difference inside the real data alone.

Probabilities of producing an overegeneralised past tense at mean age:

- Real TD: about 5.1%
- Real SLI: about 2.5%
- Synthetic SLI: about 1.4%
- Synthetic TD: about 0.1%

```r
library(dplyr)
library(ggplot2)
library(emmeans)

# Raw observed rate for plotting
age_mean <- mean(df_dev_4pop$Age_Month, na.rm = TRUE)

df_plot <- df_dev_4pop %>%
  mutate(
    Age_Month_plot = Age_c + age_mean,
    observed_rate = past_overgeneralization_errors / past_verb_opportunities,
    source_label = factor(
      source_is_synthetic,
      levels = c(0, 1),
      labels = c("Real", "Synthetic")
    ),
    clinical_group = factor(clinical_group, levels = c("TD", "SLI"))
  ) %>%
  filter(
    !is.na(observed_rate),
    !is.na(Age_Month_plot),
    !is.na(source_label),
    !is.na(clinical_group)
  )

# Prediction grid on response scale
age_grid <- seq(
  min(df_plot$Age_c, na.rm = TRUE),
  max(df_plot$Age_c, na.rm = TRUE),
  length.out = 100
)

emm_overgen <- emmeans(
  m3_overgen,
  ~ source_is_synthetic * clinical_group * Age_c,
  at = list(
    source_is_synthetic = c(0, 1),
    Age_c = age_grid
```

```r
  ),
  type = "response"
)

pred_df <- as.data.frame(emm_overgen) %>%
  mutate(
    Age_Month_plot = Age_c + age_mean,
    source_label = factor(
      source_is_synthetic,
      levels = c(0, 1),
      labels = c("Real", "Synthetic")
    ),
    clinical_group = factor(clinical_group, levels = c("TD", "SLI"))
  )

# Plot
ggplot(df_plot, aes(x = Age_Month_plot, y = observed_rate, color = source_label)) +
  geom_jitter(
    aes(size = past_verb_opportunities),
    width = 0.15, height = 0.005, alpha = 0.18
  ) +
  geom_line(
    data = pred_df,
    aes(x = Age_Month_plot, y = prob, color = source_label),
    linewidth = 1.2,
    inherit.aes = FALSE
  ) +
  geom_ribbon(
    data = pred_df,
    aes(x = Age_Month_plot, ymin = asymp.LCL, ymax = asymp.UCL, fill = source_label),
    alpha = 0.12,
    color = NA,
    inherit.aes = FALSE
  ) +
  facet_wrap(~ clinical_group) +
  labs(
    x = "Age (months)",
    y = "Predicted probability of past overgeneralization",
    color = "Source",
    fill = "Source",
    size = "Past opportunities"
  ) +
  theme_classic()
```



The synthetic data strongly under-produce past over-generalization, and the age pattern differs from the real data, especially across TD and SLI. In the real data these errors decline with age. The pattern of synthetic data is closer to floor effect.

non-finiteness

```
library(lme4)
library(lmerTest)

m0_bare <- lmer(
  bare_marking_errors_per_word ~ source_is_synthetic * clinical_group +
    factor(round_id) + (1 | cluster_id),
  data = df_4pop,
  REML = FALSE
) # Does nonfiniteness differ by source, by clinical group, and by their interaction?

m1_bare <- lmer(
  bare_marking_errors_per_word ~ source_is_synthetic * clinical_group +
    Age_c + factor(round_id) + (1 | cluster_id),
  data = df_4pop,
  REML = FALSE
)

m2_bare <- lmer(
  bare_marking_errors_per_word ~ source_is_synthetic * clinical_group +
```

```r
    Age_c +
    source_is_synthetic:Age_c +
    clinical_group:Age_c +
    factor(round_id) + (1 | cluster_id),
  data = df_4pop,
  REML = FALSE
)

m3_bare <- lmer(
  bare_marking_errors_per_word ~ source_is_synthetic * clinical_group * Age_c +
    factor(round_id) + (1 | cluster_id),
  data = df_4pop,
  REML = FALSE
)

anova(m0_bare, m1_bare, m2_bare, m3_bare)
```

```
## Data: df_4pop
## Models:
## m0_bare: bare_marking_errors_per_word ~ source_is_synthetic * clinical_group + factor(round_id) + (1
## m1_bare: bare_marking_errors_per_word ~ source_is_synthetic * clinical_group + Age_c + factor(round_
## m2_bare: bare_marking_errors_per_word ~ source_is_synthetic * clinical_group + Age_c + source_is_syn
## m3_bare: bare_marking_errors_per_word ~ source_is_synthetic * clinical_group * Age_c + factor(round_
##         npar     AIC     BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
## m0_bare    9 -2384.6 -2348.4 1201.3   -2402.6
## m1_bare   10 -2383.0 -2342.7 1201.5   -2403.0 0.3881  1     0.5333
## m2_bare   12 -2383.4 -2335.1 1203.7   -2407.4 4.3987  2     0.1109
## m3_bare   13 -2381.4 -2329.1 1203.7   -2407.4 0.0482  1     0.8262
```

```r
summary(m0_bare)
```

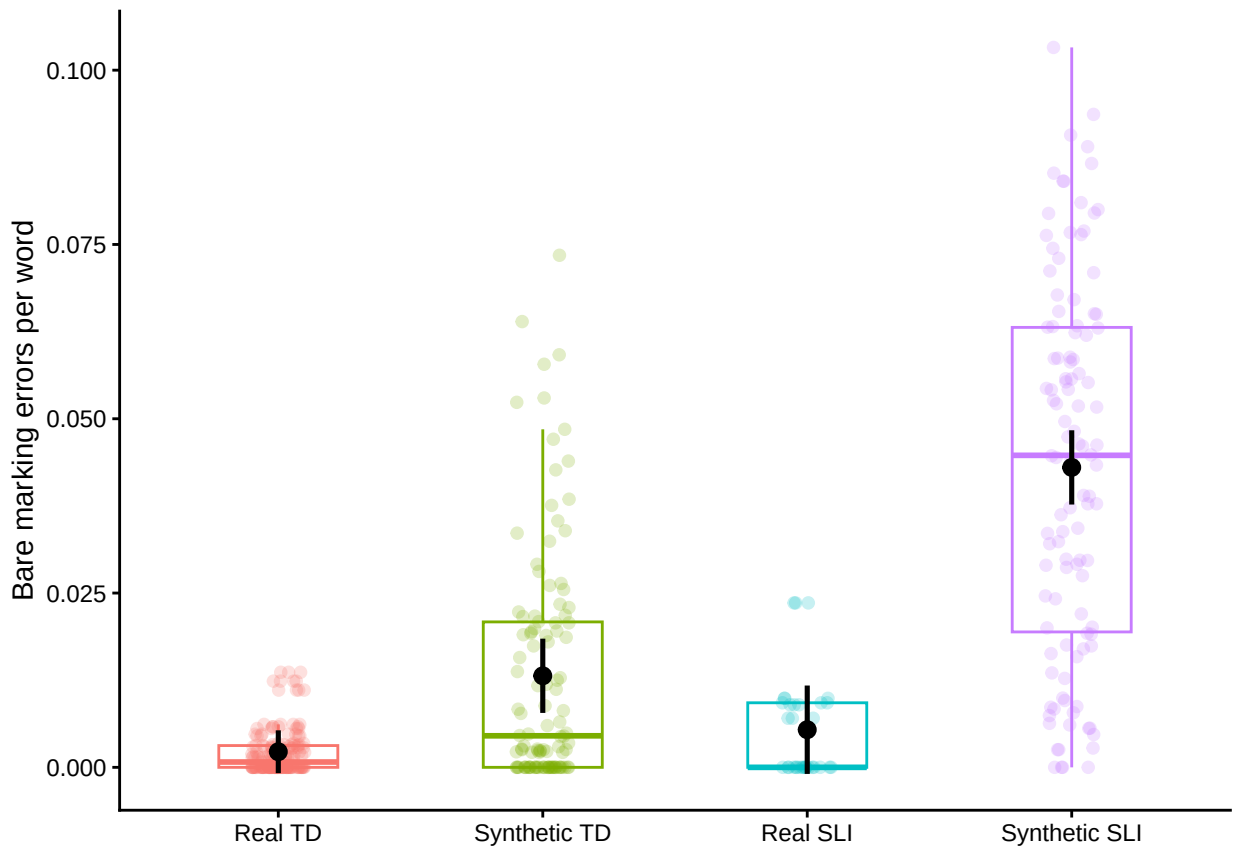
```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula: bare_marking_errors_per_word ~ source_is_synthetic * clinical_group +
##     factor(round_id) + (1 | cluster_id)
##    Data: df_4pop
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##   -2384.6   -2348.4    1201.3   -2402.6       405
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.9481 -0.1984 -0.0173  0.2652  3.7694
##
## Random effects:
##  Groups     Name        Variance  Std.Dev.
##  cluster_id (Intercept) 5.745e-05 0.007579
##  Residual               1.490e-04 0.012205
## Number of obs: 414, groups:  cluster_id, 65
##
## Fixed effects:
##                                          Estimate Std. Error         df t value
## (Intercept)                             1.944e-03  2.087e-03  2.313e+02   0.932
```

```
## source_is_synthetic                    1.094e-02  2.978e-03  7.385e+01   3.673
## clinical_groupSLI                      3.146e-03  3.465e-03  1.277e+02   0.908
## factor(round_id)round_02               1.726e-03  2.353e-03  4.114e+02   0.734
## factor(round_id)round_03              -2.506e-04  2.353e-03  4.114e+02  -0.106
## factor(round_id)round_04              -3.888e-04  2.355e-03  4.140e+02  -0.165
## source_is_synthetic:clinical_groupSLI  2.673e-02  3.836e-03  1.763e+02   6.969
##                                       Pr(>|t|)
## (Intercept)                           0.352530
## source_is_synthetic                   0.000451 ***
## clinical_groupSLI                     0.365651
## factor(round_id)round_02              0.463598
## factor(round_id)round_03              0.915245
## factor(round_id)round_04              0.868953
## source_is_synthetic:clinical_groupSLI 6.14e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) src_s_ cl_SLI f(_)_02 f(_)_03 f(_)_04
## src_s_synth -0.376
## clncl_grSLI -0.312  0.219
## fctr(r_)_02 -0.564  0.000  0.000
## fctr(r_)_03 -0.564  0.000  0.000  0.500
## fctr(r_)_04 -0.568  0.039 -0.008  0.500   0.500
## src_s_:_SLI  0.281 -0.317 -0.903  0.000   0.000   0.007
```

Main differences are about groups not developmental slopes. Bare marking errors are low in the real data, but much higher in the synthetic data, especially for SLI. The generation model appears to overgenerate nonfinite/bare-marked words, and it does so especially strongly for SLI.

```r
library(dplyr)
library(ggplot2)
library(emmeans)

emm_bare <- emmeans(
  m1_bare,
  ~ source_is_synthetic * clinical_group,
  at = list(source_is_synthetic = c(0, 1))
)

emm_bare_df <- as.data.frame(emm_bare) %>%
  mutate(
    population = factor(
      paste(ifelse(source_is_synthetic == 1, "Synthetic", "Real"), clinical_group),
      levels = c("Real TD", "Synthetic TD", "Real SLI", "Synthetic SLI")
    )
  )

df_plot <- df_4pop %>%
  mutate(
    population = factor(
      paste(ifelse(source_is_synthetic == 1, "Synthetic", "Real"), clinical_group),
      levels = c("Real TD", "Synthetic TD", "Real SLI", "Synthetic SLI")
    )
```

```
  ) %>%
  filter(!is.na(bare_marking_errors_per_word))

ggplot(df_plot, aes(x = population, y = bare_marking_errors_per_word, color = population)) +
  geom_boxplot(width = 0.45, alpha = 0.15, outlier.shape = NA) +
  geom_jitter(width = 0.10, alpha = 0.22, size = 1.7) +
  geom_pointrange(
    data = emm_bare_df,
    aes(x = population, y = emmean, ymin = lower.CL, ymax = upper.CL),
    inherit.aes = FALSE,
    linewidth = 0.9,
    color = "black"
  ) +
  labs(
    x = NULL,
    y = "Bare marking errors per word"
  ) +
  theme_classic() +
  theme(legend.position = "none")
```



agreement

```
m0_agr <- lmer(
  agreement_errors_per_word ~ source_is_synthetic * clinical_group +
    factor(round_id) + (1 | cluster_id),
  data = df_4pop,
```

```r
  REML = FALSE
)

m1_agr <- lmer(
  agreement_errors_per_word ~ source_is_synthetic * clinical_group +
    Age_c + factor(round_id) + (1 | cluster_id),
  data = df_4pop,
  REML = FALSE
)

m2_agr <- lmer(
  agreement_errors_per_word ~ source_is_synthetic * clinical_group +
    Age_c +
    source_is_synthetic:Age_c +
    clinical_group:Age_c +
    factor(round_id) + (1 | cluster_id),
  data = df_4pop,
  REML = FALSE
)

m3_agr <- lmer(
  agreement_errors_per_word ~ source_is_synthetic * clinical_group * Age_c +
    factor(round_id) + (1 | cluster_id),
  data = df_4pop,
  REML = FALSE
)

anova(m0_agr, m1_agr, m2_agr, m3_agr)
```

```
## Data: df_4pop
## Models:
## m0_agr: agreement_errors_per_word ~ source_is_synthetic * clinical_group + factor(round_id) + (1 | cl
## m1_agr: agreement_errors_per_word ~ source_is_synthetic * clinical_group + Age_c + factor(round_id) +
## m2_agr: agreement_errors_per_word ~ source_is_synthetic * clinical_group + Age_c + source_is_synthetic
## m3_agr: agreement_errors_per_word ~ source_is_synthetic * clinical_group * Age_c + factor(round_id) +
##        npar     AIC     BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
## m0_agr    9 -2334.1 -2297.8 1176.0   -2352.1
## m1_agr   10 -2332.7 -2292.4 1176.3   -2352.7 0.5892  1     0.4427
## m2_agr   12 -2332.4 -2284.1 1178.2   -2356.4 3.8026  2     0.1494
## m3_agr   13 -2331.3 -2279.0 1178.7   -2357.3 0.8893  1     0.3457
```

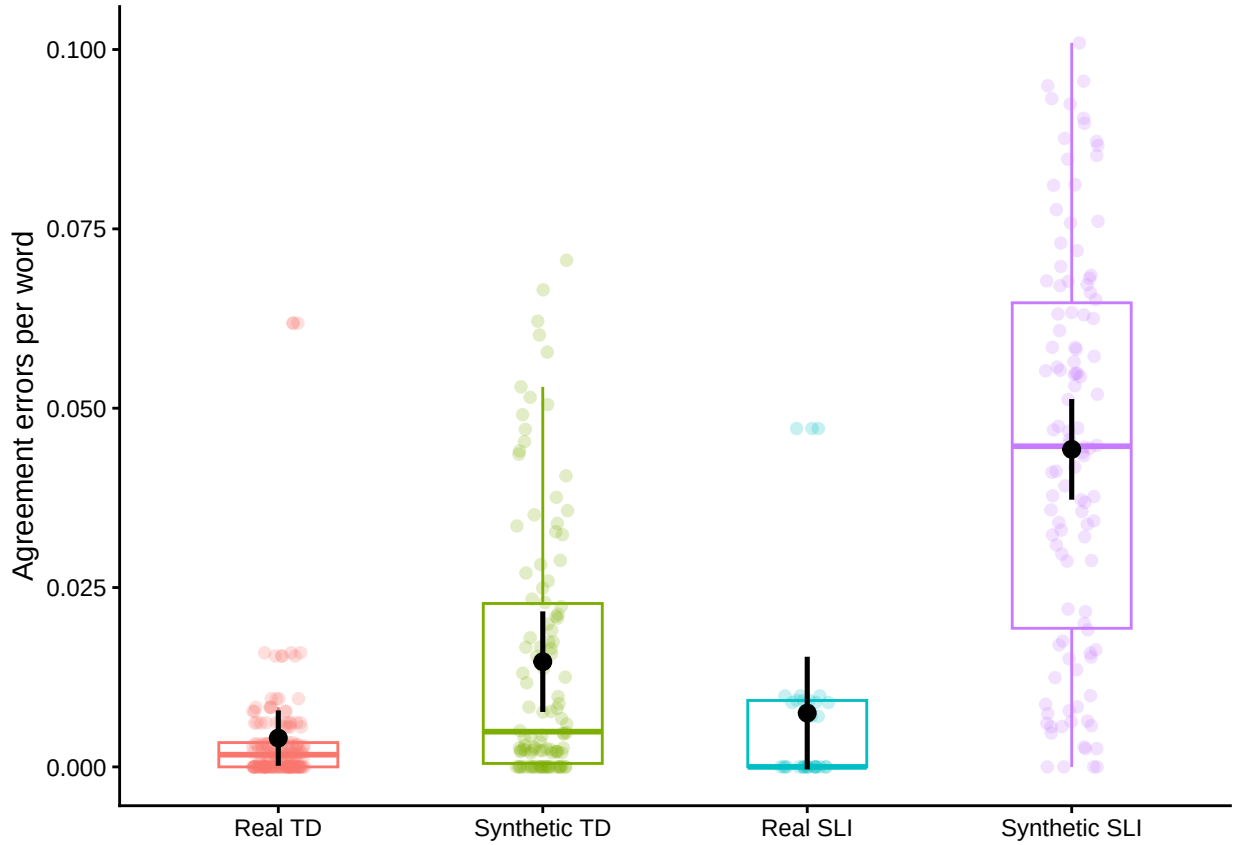
```r
summary(m0_agr)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula: agreement_errors_per_word ~ source_is_synthetic * clinical_group +
##     factor(round_id) + (1 | cluster_id)
##    Data: df_4pop
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##   -2334.1   -2297.8    1176.0   -2352.1       405
##
```

```
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.7963 -0.1916 -0.0418  0.2706  4.0251
##
## Random effects:
##  Groups     Name        Variance  Std.Dev.
##  cluster_id (Intercept) 0.0001111 0.01054
##  Residual               0.0001576 0.01256
## Number of obs: 414, groups:  cluster_id, 65
##
## Fixed effects:
##                                        Estimate Std. Error         df t value
## (Intercept)                           3.558e-03  2.432e-03  1.685e+02   1.463
## source_is_synthetic                   1.065e-02  3.890e-03  5.788e+01   2.739
## clinical_groupSLI                     3.487e-03  4.296e-03  8.548e+01   0.812
## factor(round_id)round_02              1.678e-03  2.496e-03  4.077e+02   0.672
## factor(round_id)round_03              2.216e-04  2.496e-03  4.077e+02   0.089
## factor(round_id)round_04             -6.897e-05  2.484e-03  4.015e+02  -0.028
## source_is_synthetic:clinical_groupSLI  2.611e-02  4.617e-03  1.120e+02   5.655
##                                      Pr(>|t|)
## (Intercept)                           0.14543
## source_is_synthetic                   0.00819 **
## clinical_groupSLI                     0.41923
## factor(round_id)round_02              0.50183
## factor(round_id)round_03              0.92930
## factor(round_id)round_04              0.97786
## source_is_synthetic:clinical_groupSLI 1.21e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) src_s_ cl_SLI f(_)_02 f(_)_03 f(_)_04
## src_s_synth -0.386
## clncl_grSLI -0.340  0.214
## fctr(r_)_02 -0.513  0.000  0.000
## fctr(r_)_03 -0.513  0.000  0.000  0.500
## fctr(r_)_04 -0.517  0.031 -0.007  0.502   0.502
## src_s_:_SLI  0.317 -0.278 -0.930  0.000   0.000   0.006
```

```r
library(dplyr)
library(ggplot2)
library(emmeans)

# Adjusted means from the retained model
emm_agr <- emmeans(
  m0_agr,
  ~ source_is_synthetic * clinical_group,
  at = list(source_is_synthetic = c(0, 1))
)

emm_agr_df <- as.data.frame(emm_agr) %>%
  mutate(
    population = factor(
      paste(ifelse(source_is_synthetic == 1, "Synthetic", "Real"), clinical_group),
```

```r
      levels = c("Real TD", "Synthetic TD", "Real SLI", "Synthetic SLI")
    )
  )

# Raw data
df_plot <- df_4pop %>%
  mutate(
    population = factor(
      paste(ifelse(source_is_synthetic == 1, "Synthetic", "Real"), clinical_group),
      levels = c("Real TD", "Synthetic TD", "Real SLI", "Synthetic SLI")
    )
  ) %>%
  filter(!is.na(agreement_errors_per_word), !is.na(population))

# Plot
ggplot(df_plot, aes(x = population, y = agreement_errors_per_word, color = population)) +
  geom_boxplot(width = 0.45, alpha = 0.15, outlier.shape = NA) +
  geom_jitter(width = 0.10, alpha = 0.22, size = 1.7) +
  geom_pointrange(
    data = emm_agr_df,
    aes(x = population, y = emmean, ymin = lower.CL, ymax = upper.CL),
    inherit.aes = FALSE,
    linewidth = 0.9,
    color = "black"
  ) +
  labs(
    x = NULL,
    y = "Agreement errors per word"
  ) +
  theme_classic() +
  theme(legend.position = "none")
```



The synthetic data inflate morphosyntactic error rates, and the inflation is especially strong for SLI.

Usage Measures

auxiliaries

```
m0_auxcop <- lmer(
  Syntax_AuxCop_Utt_Prop ~ source_is_synthetic * clinical_group +
    factor(round_id) + (1 | cluster_id),
  data = df_4pop,
  REML = FALSE
)

m1_auxcop <- lmer(
  Syntax_AuxCop_Utt_Prop ~ source_is_synthetic * clinical_group +
    Age_c + factor(round_id) + (1 | cluster_id),
  data = df_4pop,
  REML = FALSE
)

m2_auxcop <- lmer(
  Syntax_AuxCop_Utt_Prop ~ source_is_synthetic * clinical_group +
    Age_c +
    source_is_synthetic:Age_c +
    clinical_group:Age_c +
```

```r
    factor(round_id) + (1 | cluster_id),
  data = df_4pop,
  REML = FALSE
)

m3_auxcop <- lmer(
  Syntax_AuxCop_Utt_Prop ~ source_is_synthetic * clinical_group * Age_c +
    factor(round_id) + (1 | cluster_id),
  data = df_4pop,
  REML = FALSE
)

anova(m0_auxcop, m1_auxcop, m2_auxcop, m3_auxcop)
```

```
## Data: df_4pop
## Models:
## m0_auxcop: Syntax_AuxCop_Utt_Prop ~ source_is_synthetic * clinical_group + factor(round_id) + (1 | cl
## m1_auxcop: Syntax_AuxCop_Utt_Prop ~ source_is_synthetic * clinical_group + Age_c + factor(round_id) +
## m2_auxcop: Syntax_AuxCop_Utt_Prop ~ source_is_synthetic * clinical_group + Age_c + source_is_synthetic
## m3_auxcop: Syntax_AuxCop_Utt_Prop ~ source_is_synthetic * clinical_group * Age_c + factor(round_id) +
##           npar     AIC     BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
## m0_auxcop    9 -793.85 -757.62 405.92   -811.85
## m1_auxcop   10 -795.68 -755.42 407.84   -815.68 3.8352  1    0.05019 .
## m2_auxcop   12 -797.06 -748.75 410.53   -821.06 5.3767  2    0.06799 .
## m3_auxcop   13 -795.14 -742.81 410.57   -821.14 0.0841  1    0.77184
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

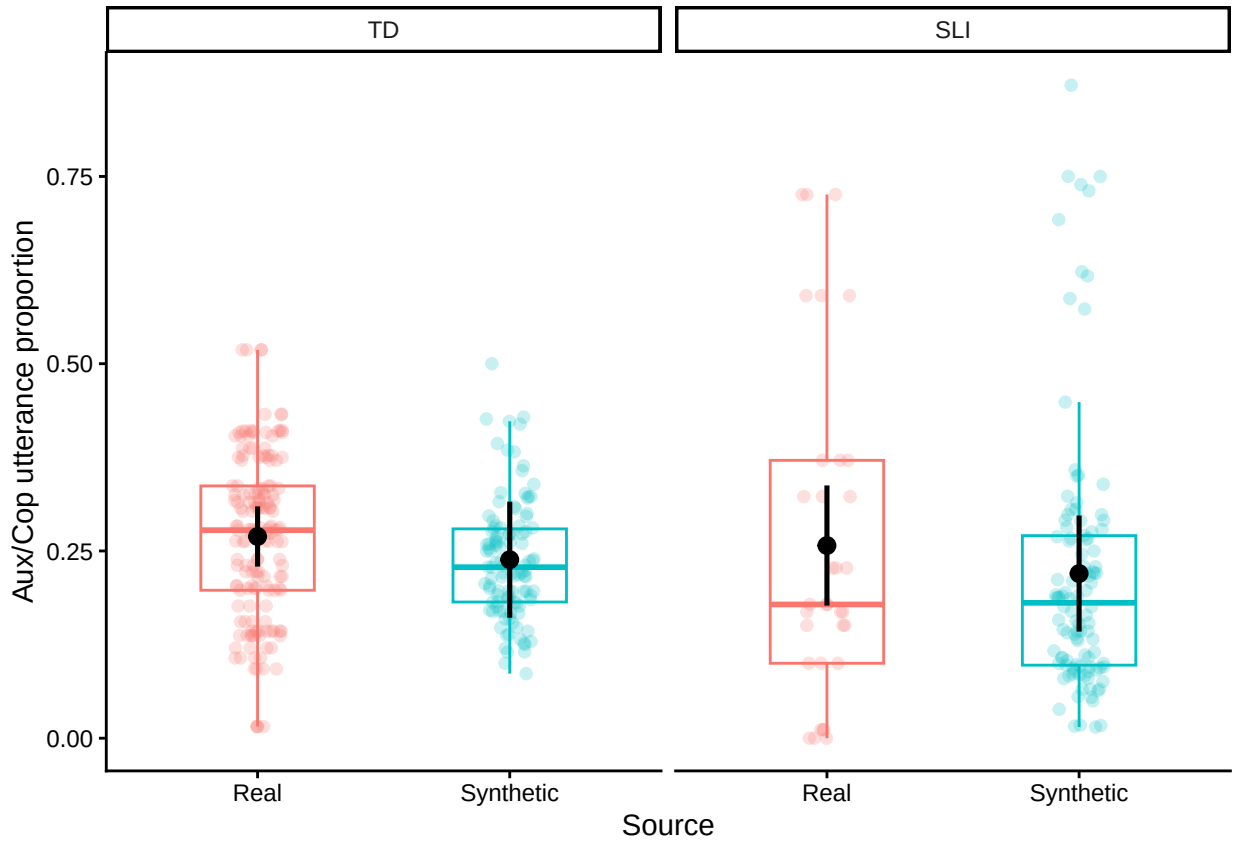
```r
summary(m1_auxcop)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula: Syntax_AuxCop_Utt_Prop ~ source_is_synthetic * clinical_group +
##     Age_c + factor(round_id) + (1 | cluster_id)
##    Data: df_4pop
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    -795.7    -755.4     407.8    -815.7       404
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.9614 -0.2187 -0.0053  0.2520  5.1552
##
## Random effects:
##  Groups     Name        Variance Std.Dev.
##  cluster_id (Intercept) 0.014909 0.12210
##  Residual               0.005371 0.07329
## Number of obs: 414, groups:  cluster_id, 65
##
## Fixed effects:
##                                          Estimate Std. Error         df t value
## (Intercept)                             2.688e-01  2.159e-02  9.513e+01  12.455
```

```
## source_is_synthetic                   -3.095e-02  4.227e-02  5.918e+01  -0.732
## clinical_groupSLI                     -1.204e-02  4.358e-02  6.696e+01  -0.276
## Age_c                                  1.193e-02  6.032e-03  6.535e+01   1.978
## factor(round_id)round_02               2.294e-04  1.507e-02  3.668e+02   0.015
## factor(round_id)round_03               1.368e-03  1.508e-02  3.667e+02   0.091
## factor(round_id)round_04               2.238e-04  1.491e-02  3.630e+02   0.015
## source_is_synthetic:clinical_groupSLI -6.477e-03  4.468e-02  7.398e+01  -0.145
##                                       Pr(>|t|)
## (Intercept)                             <2e-16 ***
## source_is_synthetic                     0.4669
## clinical_groupSLI                       0.7832
## Age_c                                   0.0522 .
## factor(round_id)round_02                0.9879
## factor(round_id)round_03                0.9278
## factor(round_id)round_04                0.9880
## source_is_synthetic:clinical_groupSLI   0.8851
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) src_s_ cl_SLI Age_c  f(_)_02 f(_)_03 f(_)_04
## src_s_synth -0.420
## clncl_grSLI -0.404  0.206
## Age_c       -0.012  0.023 -0.008
## fctr(r_)_02 -0.349  0.000  0.000 -0.008
## fctr(r_)_03 -0.349  0.000  0.000 -0.015  0.500
## fctr(r_)_04 -0.352  0.017 -0.004 -0.018  0.506   0.506
## src_s_:_SLI  0.394 -0.227 -0.975  0.003  0.000   0.000   0.004
```

```r
library(dplyr)
library(ggplot2)
library(emmeans)

# Adjusted means from the primary model
emm_auxcop <- emmeans(
  m1_auxcop,
  ~ source_is_synthetic * clinical_group,
  at = list(source_is_synthetic = c(0, 1))
)

emm_auxcop_df <- as.data.frame(emm_auxcop) %>%
  mutate(
    source_label = factor(
      source_is_synthetic,
      levels = c(0, 1),
      labels = c("Real", "Synthetic")
    ),
    clinical_group = factor(clinical_group, levels = c("TD", "SLI"))
  )

# Raw data
df_plot <- df_4pop %>%
  mutate(
    source_label = factor(
```

```r
      source_is_synthetic,
      levels = c(0, 1),
      labels = c("Real", "Synthetic")
    ),
    clinical_group = factor(clinical_group, levels = c("TD", "SLI"))
  ) %>%
  filter(!is.na(Syntax_AuxCop_Utt_Prop))

# Plot
ggplot(df_plot, aes(x = source_label, y = Syntax_AuxCop_Utt_Prop, color = source_label)) +
  geom_boxplot(width = 0.45, alpha = 0.15, outlier.shape = NA) +
  geom_jitter(width = 0.10, alpha = 0.22, size = 1.7) +
  geom_pointrange(
    data = emm_auxcop_df,
    aes(x = source_label, y = emmean, ymin = lower.CL, ymax = upper.CL),
    inherit.aes = FALSE,
    linewidth = 0.9,
    color = "black"
  ) +
  facet_wrap(~ clinical_group) +
  labs(
    x = "Source",
    y = "Aux/Cop utterance proportion"
  ) +
  theme_classic() +
  theme(legend.position = "none")
```



Looks consistently similar.

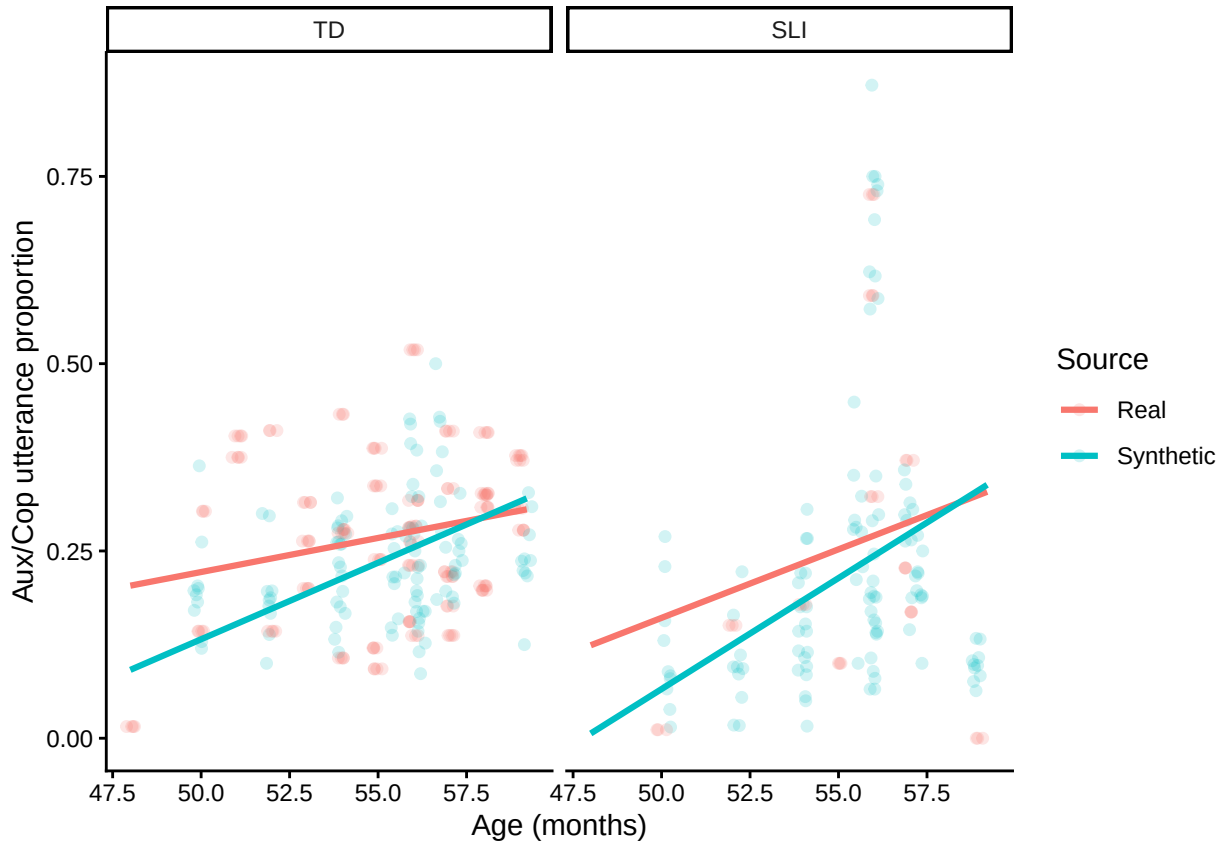
```
summary(m2_auxcop)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula: Syntax_AuxCop_Utt_Prop ~ source_is_synthetic * clinical_group +
##     Age_c + source_is_synthetic:Age_c + clinical_group:Age_c +
##     factor(round_id) + (1 | cluster_id)
##    Data: df_4pop
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    -797.1    -748.7     410.5    -821.1       402
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.9394 -0.2169 -0.0151  0.2093  5.1355
##
## Random effects:
##  Groups     Name        Variance Std.Dev.
##  cluster_id (Intercept) 0.014979 0.1224
##  Residual               0.005286 0.0727
## Number of obs: 414, groups:  cluster_id, 65
##
## Fixed effects:
##                                          Estimate Std. Error         df t value
```

```
## (Intercept)                            2.699e-01  2.162e-02  9.394e+01  12.484
## source_is_synthetic                   -3.039e-02  4.236e-02  5.802e+01  -0.717
## clinical_groupSLI                     -1.305e-02  4.364e-02  6.554e+01  -0.299
## Age_c                                  9.099e-03  6.656e-03  6.507e+01   1.367
## factor(round_id)round_02              -5.721e-04  1.500e-02  3.616e+02  -0.038
## factor(round_id)round_03               2.066e-04  1.503e-02  3.583e+02   0.014
## factor(round_id)round_04              -1.104e-03  1.489e-02  3.515e+02  -0.074
## source_is_synthetic:clinical_groupSLI -5.292e-03  4.473e-02  7.234e+01  -0.118
## source_is_synthetic:Age_c              1.134e-02  1.639e-02  6.392e+01   0.692
## clinical_groupSLI:Age_c                9.196e-03  4.035e-03  3.983e+02   2.279
##                                       Pr(>|t|)
## (Intercept)                             <2e-16 ***
## source_is_synthetic                     0.4760
## clinical_groupSLI                       0.7658
## Age_c                                   0.1763
## factor(round_id)round_02                0.9696
## factor(round_id)round_03                0.9890
## factor(round_id)round_04                0.9410
## source_is_synthetic:clinical_groupSLI   0.9062
## source_is_synthetic:Age_c               0.4914
## clinical_groupSLI:Age_c                 0.0232 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) src_s_ cl_SLI Age_c  f(_)_02 f(_)_03 f(_)_04 s__:_S s__:A_
## src_s_synth -0.419
## clncl_grSLI -0.404  0.206
## Age_c       -0.035  0.011 -0.008
## fctr(r_)_02 -0.349 -0.002  0.000  0.023
## fctr(r_)_03 -0.350 -0.003  0.000  0.027  0.503
## fctr(r_)_04 -0.353  0.014 -0.004  0.032  0.509   0.511
## src_s_:_SLI  0.393 -0.227 -0.976  0.008  0.001   0.001   0.005
## src_s_sy:A_  0.058  0.025  0.003 -0.415 -0.073  -0.097  -0.115  -0.014
## clnc_SLI:A_  0.009  0.001 -0.011 -0.103 -0.008  -0.013  -0.015   0.015  0.096
```

```r
age_mean <- mean(df_4pop$Age_Month, na.rm = TRUE)

df_age_plot <- df_4pop %>%
  mutate(
    Age_Month_plot = Age_c + age_mean,
    source_label = factor(
      source_is_synthetic,
      levels = c(0, 1),
      labels = c("Real", "Synthetic")
    ),
    clinical_group = factor(clinical_group, levels = c("TD", "SLI"))
  ) %>%
  filter(
    !is.na(Syntax_AuxCop_Utt_Prop),
    !is.na(Age_c),
    !is.na(Age_Month_plot)
  )
```

```r
age_grid <- seq(
  min(df_age_plot$Age_c, na.rm = TRUE),
  max(df_age_plot$Age_c, na.rm = TRUE),
  length.out = 100
)

emm_age <- emmeans(
  m2_auxcop,
  ~ source_is_synthetic * clinical_group * Age_c,
  at = list(
    source_is_synthetic = c(0, 1),
    Age_c = age_grid
  )
)

pred_df <- as.data.frame(emm_age) %>%
  mutate(
    Age_Month_plot = Age_c + age_mean,
    source_label = factor(
      source_is_synthetic,
      levels = c(0, 1),
      labels = c("Real", "Synthetic")
    ),
    clinical_group = factor(clinical_group, levels = c("TD", "SLI"))
  )

ggplot(df_age_plot, aes(x = Age_Month_plot, y = Syntax_AuxCop_Utt_Prop, color = source_label)) +
  geom_jitter(width = 0.15, height = 0, alpha = 0.18, size = 1.6) +
  geom_line(
    data = pred_df,
    aes(x = Age_Month_plot, y = emmean, color = source_label),
    linewidth = 1.1,
    inherit.aes = FALSE
  ) +
  facet_wrap(~ clinical_group) +
  labs(
    x = "Age (months)",
    y = "Aux/Cop utterance proportion",
    color = "Source"
  ) +
  theme_classic()
```



Looks consistently similar.

determiners

```
df_dev_4pop <- df_dev_4pop %>%
  mutate(
    det_art_per_utt = `det:art` / `#utterances`
  )
```

```
m0_detuse <- lmer(
  det_art_per_utt ~ source_is_synthetic * clinical_group +
    factor(round_id) + (1 | cluster_id),
  data = df_dev_4pop,
  REML = FALSE
)

m1_detuse <- lmer(
  det_art_per_utt ~ source_is_synthetic * clinical_group +
    Age_c + factor(round_id) + (1 | cluster_id),
  data = df_dev_4pop,
  REML = FALSE
)

m2_detuse <- lmer(
  det_art_per_utt ~ source_is_synthetic * clinical_group +
```

```r
    Age_c +
    source_is_synthetic:Age_c +
    clinical_group:Age_c +
    factor(round_id) + (1 | cluster_id),
  data = df_dev_4pop,
  REML = FALSE
)

m3_detuse <- lmer(
  det_art_per_utt ~ source_is_synthetic * clinical_group * Age_c +
    factor(round_id) + (1 | cluster_id),
  data = df_dev_4pop,
  REML = FALSE
)

anova(m0_detuse, m1_detuse, m2_detuse, m3_detuse)
```

```
## Data: df_dev_4pop
## Models:
## m0_detuse: det_art_per_utt ~ source_is_synthetic * clinical_group + factor(round_id) + (1 | cluster_i
## m1_detuse: det_art_per_utt ~ source_is_synthetic * clinical_group + Age_c + factor(round_id) + (1 | c
## m2_detuse: det_art_per_utt ~ source_is_synthetic * clinical_group + Age_c + source_is_synthetic:Age_c
## m3_detuse: det_art_per_utt ~ source_is_synthetic * clinical_group * Age_c + factor(round_id) + (1 | c
##           npar     AIC     BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
## m0_detuse    9 -369.50 -333.31 193.75   -387.50
## m1_detuse   10 -368.10 -327.89 194.05   -388.10 0.5956  1    0.44027
## m2_detuse   12 -369.05 -320.80 196.53   -393.05 4.9499  2    0.08417 .
## m3_detuse   13 -368.98 -316.71 197.49   -394.98 1.9292  1    0.16484
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
summary(m0_detuse)
```

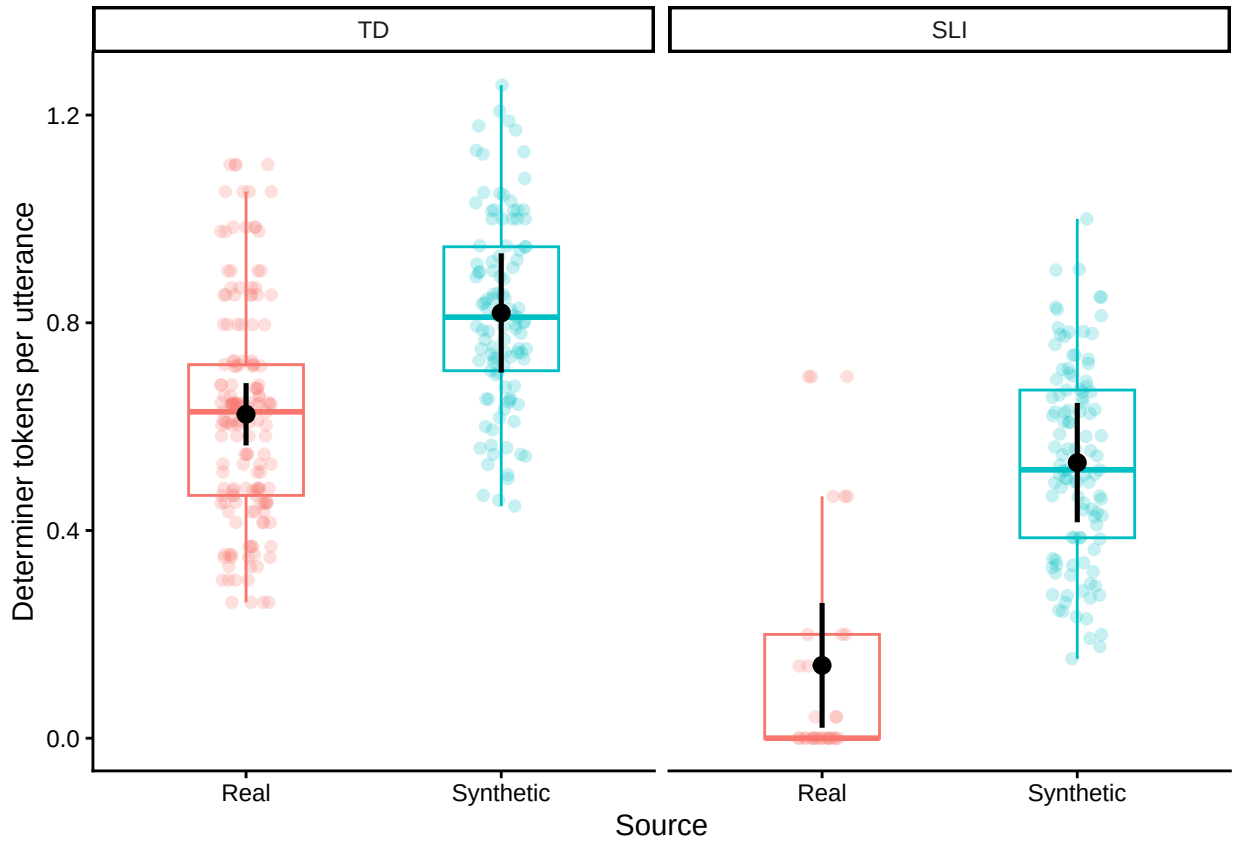
```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula:
## det_art_per_utt ~ source_is_synthetic * clinical_group + factor(round_id) +
##     (1 | cluster_id)
##    Data: df_dev_4pop
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    -369.5    -333.3     193.8    -387.5       403
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.8938 -0.2942 -0.0209  0.2811  3.6888
##
## Random effects:
##  Groups     Name        Variance Std.Dev.
##  cluster_id (Intercept) 0.03284  0.1812
##  Residual               0.01562  0.1250
## Number of obs: 412, groups:  cluster_id, 65
##
```

```
## Fixed effects:
##                                          Estimate Std. Error         df t value
## (Intercept)                             6.219e-01  3.329e-02  1.038e+02  18.679
## source_is_synthetic                     1.951e-01  6.320e-02  5.777e+01   3.088
## clinical_groupSLI                      -4.835e-01  6.574e-02  6.770e+01  -7.355
## factor(round_id)round_02                2.391e-03  2.560e-02  3.703e+02   0.093
## factor(round_id)round_03                4.451e-03  2.560e-02  3.703e+02   0.174
## factor(round_id)round_04                7.616e-04  2.533e-02  3.657e+02   0.030
## source_is_synthetic:clinical_groupSLI  1.953e-01  6.788e-02  7.691e+01   2.878
##                                        Pr(>|t|)
## (Intercept)                             < 2e-16 ***
## source_is_synthetic                     0.00310 **
## clinical_groupSLI                      3.31e-10 ***
## factor(round_id)round_02                0.92566
## factor(round_id)round_03                0.86208
## factor(round_id)round_04                0.97603
## source_is_synthetic:clinical_groupSLI  0.00518 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) src_s_ cl_SLI f(_)_02 f(_)_03 f(_)_04
## src_s_synth -0.414
## clncl_grSLI -0.393  0.207
## fctr(r_)_02 -0.385  0.000  0.000
## fctr(r_)_03 -0.385  0.000  0.000  0.500
## fctr(r_)_04 -0.387  0.019 -0.004  0.505   0.505
## src_s_:_SLI  0.381 -0.234 -0.968  0.000   0.000   0.004
```

Synthetic transcripts use more determiners per utterance than real transcripts SLI transcripts use fewer determiners per utterance than TD transcripts the synthetic boost is larger in SLI than in TD

```r
library(dplyr)
library(ggplot2)
library(emmeans)

# Create emmeans object from the retained model
emm_detuse <- emmeans(
  m0_detuse,
  ~ source_is_synthetic * clinical_group,
  at = list(source_is_synthetic = c(0, 1))
)

emm_detuse_df <- as.data.frame(emm_detuse) %>%
  mutate(
    source_label = factor(
      source_is_synthetic,
      levels = c(0, 1),
      labels = c("Real", "Synthetic")
    ),
    clinical_group = factor(clinical_group, levels = c("TD", "SLI"))
  )

df_plot <- df_dev_4pop %>%
```

```r
  mutate(
    source_label = factor(
      source_is_synthetic,
      levels = c(0, 1),
      labels = c("Real", "Synthetic")
    ),
    clinical_group = factor(clinical_group, levels = c("TD", "SLI"))
  ) %>%
  filter(!is.na(det_art_per_utt))

ggplot(df_plot, aes(x = source_label, y = det_art_per_utt, color = source_label)) +
  geom_boxplot(width = 0.45, alpha = 0.15, outlier.shape = NA) +
  geom_jitter(width = 0.10, alpha = 0.22, size = 1.7) +
  geom_pointrange(
    data = emm_detuse_df,
    aes(x = source_label, y = emmean, ymin = lower.CL, ymax = upper.CL),
    inherit.aes = FALSE,
    linewidth = 0.9,
    color = "black"
  ) +
  facet_wrap(~ clinical_group) +
  labs(
    x = "Source",
    y = "Determiner tokens per utterance"
  ) +
  theme_classic() +
  theme(legend.position = "none")
```




```r
summary(m2_detuse)
```

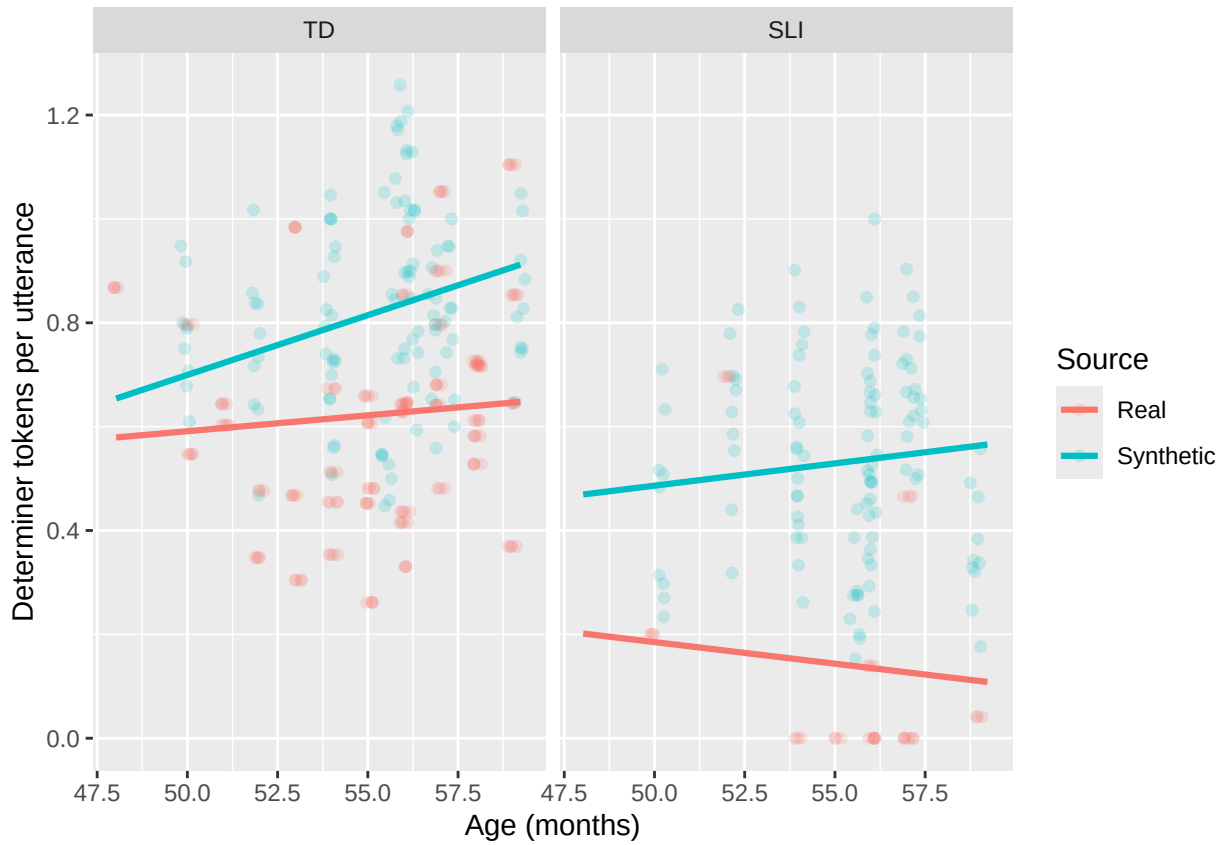
```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula: det_art_per_utt ~ source_is_synthetic * clinical_group + Age_c +
##     source_is_synthetic:Age_c + clinical_group:Age_c + factor(round_id) +
##     (1 | cluster_id)
##    Data: df_dev_4pop
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    -369.0    -320.8     196.5    -393.0       400
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.7508 -0.2972 -0.0217  0.2847  3.6891
##
## Random effects:
##  Groups     Name        Variance Std.Dev.
##  cluster_id (Intercept) 0.03193  0.1787
##  Residual               0.01545  0.1243
## Number of obs: 412, groups:  cluster_id, 65
##
## Fixed effects:
##                                          Estimate Std. Error         df t value
## (Intercept)                             6.230e-01  3.300e-02  1.054e+02  18.877
```

```
## source_is_synthetic                    1.973e-01  6.238e-02  5.699e+01   3.163
## clinical_groupSLI                     -4.820e-01  6.490e-02  6.702e+01  -7.427
## Age_c                                  6.117e-03  9.897e-03  6.633e+01   0.618
## factor(round_id)round_02               8.078e-04  2.555e-02  3.650e+02   0.032
## factor(round_id)round_03               2.400e-03  2.563e-02  3.607e+02   0.094
## factor(round_id)round_04              -1.729e-03  2.542e-02  3.516e+02  -0.068
## source_is_synthetic:clinical_groupSLI  1.928e-01  6.706e-02  7.629e+01   2.876
## source_is_synthetic:Age_c              1.690e-02  2.429e-02  6.176e+01   0.696
## clinical_groupSLI:Age_c               -1.444e-02  6.946e-03  3.994e+02  -2.079
##                                       Pr(>|t|)
## (Intercept)                            < 2e-16 ***
## source_is_synthetic                    0.00251 **
## clinical_groupSLI                     2.59e-10 ***
## Age_c                                  0.53865
## factor(round_id)round_02               0.97479
## factor(round_id)round_03               0.92543
## factor(round_id)round_04               0.94583
## source_is_synthetic:clinical_groupSLI  0.00523 **
## source_is_synthetic:Age_c              0.48910
## clinical_groupSLI:Age_c                0.03827 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) src_s_ cl_SLI Age_c  f(_)_02 f(_)_03 f(_)_04 s__:_S s__:A_
## src_s_synth -0.410
## clncl_grSLI -0.391  0.207
## Age_c       -0.039  0.011 -0.008
## fctr(r_)_02 -0.390 -0.002  0.000  0.026
## fctr(r_)_03 -0.391 -0.003  0.000  0.031  0.505
## fctr(r_)_04 -0.395  0.016 -0.005  0.036  0.510   0.513
## src_s_:_SLI  0.378 -0.235 -0.968  0.007  0.001   0.002   0.006
## src_s_sy:A_  0.073  0.024  0.003 -0.413 -0.085  -0.113  -0.134  -0.015
## clnc_SLI:A_  0.008  0.001 -0.013 -0.118 -0.007  -0.011  -0.013   0.013  0.057
```

Determiner usage per utterance was lower in SLI than TD in the real data, but synthetic transcripts used determiners more often overall, with an especially large increase for SLI. This suggests that the synthetic data inflate determiner usage and partly reduce the real clinical-group difference.

```r
library(emmeans)

age_mean <- mean(df_dev_4pop$Age_Month, na.rm = TRUE)

df_age_plot <- df_dev_4pop %>%
  mutate(
    Age_Month_plot = Age_c + age_mean,
    source_label = factor(
      source_is_synthetic,
      levels = c(0, 1),
      labels = c("Real", "Synthetic")
    ),
    clinical_group = factor(clinical_group, levels = c("TD", "SLI"))
  ) %>%
  filter(
```

```r
    !is.na(det_art_per_utt),
    !is.na(Age_c),
    !is.na(Age_Month_plot)
  )

age_grid <- seq(
  min(df_age_plot$Age_c, na.rm = TRUE),
  max(df_age_plot$Age_c, na.rm = TRUE),
  length.out = 100
)

emm_age <- emmeans(
  m2_detuse,
  ~ source_is_synthetic * clinical_group * Age_c,
  at = list(
    source_is_synthetic = c(0, 1),
    Age_c = age_grid
  )
)

pred_df <- as.data.frame(emm_age) %>%
  mutate(
    Age_Month_plot = Age_c + age_mean,
    source_label = factor(
      source_is_synthetic,
      levels = c(0, 1),
      labels = c("Real", "Synthetic")
    ),
    clinical_group = factor(clinical_group, levels = c("TD", "SLI"))
  )

ggplot(df_age_plot, aes(x = Age_Month_plot, y = det_art_per_utt, color = source_label)) +
  geom_jitter(width = 0.15, height = 0, alpha = 0.18, size = 1.6) +
  geom_line(
    data = pred_df,
    aes(x = Age_Month_plot, y = emmean, color = source_label),
    linewidth = 1.1,
    inherit.aes = FALSE
  ) +
  facet_wrap(~ clinical_group) +
  labs(
    x = "Age (months)",
    y = "Determiner tokens per utterance",
    color = "Source"
  ) #+
```




```
  theme_classic()
```

```
## <theme> List of 144
##  $ line                            : <ggplot2::element_line>
##   ..@ colour       : chr "black"
##   ..@ linewidth    : num 0.5
##   ..@ linetype     : num 1
##   ..@ lineend      : chr "butt"
##   ..@ linejoin     : chr "round"
##   ..@ arrow        : logi FALSE
##   ..@ arrow.fill   : chr "black"
##   ..@ inherit.blank: logi TRUE
##  $ rect                            : <ggplot2::element_rect>
##   ..@ fill         : chr "white"
##   ..@ colour       : chr "black"
##   ..@ linewidth    : num 0.5
##   ..@ linetype     : num 1
##   ..@ linejoin     : chr "round"
##   ..@ inherit.blank: logi TRUE
##  $ text                            : <ggplot2::element_text>
##   ..@ family       : chr ""
##   ..@ face         : chr "plain"
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
```

```
##   ..@ colour       : chr "black"
##   ..@ size         : num 11
##   ..@ hjust        : num 0.5
##   ..@ vjust        : num 0.5
##   ..@ angle        : num 0
##   ..@ lineheight   : num 0.9
##   ..@ margin       : <ggplot2::margin> num [1:4] 0 0 0 0
##   ..@ debug        : logi FALSE
##   ..@ inherit.blank: logi TRUE
##  $ title                          : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : NULL
##   ..@ vjust        : NULL
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : NULL
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ point                          : <ggplot2::element_point>
##   ..@ colour       : chr "black"
##   ..@ shape        : num 19
##   ..@ size         : num 1.5
##   ..@ fill         : chr "white"
##   ..@ stroke       : num 0.5
##   ..@ inherit.blank: logi TRUE
##  $ polygon                        : <ggplot2::element_polygon>
##   ..@ fill         : chr "white"
##   ..@ colour       : chr "black"
##   ..@ linewidth    : num 0.5
##   ..@ linetype     : num 1
##   ..@ linejoin     : chr "round"
##   ..@ inherit.blank: logi TRUE
##  $ geom                           : <ggplot2::element_geom>
##   ..@ ink        : chr "black"
##   ..@ paper      : chr "white"
##   ..@ accent     : chr "#3366FF"
##   ..@ linewidth  : num 0.5
##   ..@ borderwidth: num 0.5
##   ..@ linetype   : int 1
##   ..@ bordertype : int 1
##   ..@ family     : chr ""
##   ..@ fontsize   : num 3.87
##   ..@ pointsize  : num 1.5
##   ..@ pointshape : num 19
##   ..@ colour     : NULL
##   ..@ fill       : NULL
##  $ spacing                        : 'simpleUnit' num 5.5points
##   ..- attr(*, "unit")= int 8
```

```
##  $ margins                         : <ggplot2::margin> num [1:4] 5.5 5.5 5.5 5.5
##  $ aspect.ratio                    : NULL
##  $ axis.title                      : NULL
##  $ axis.title.x                    : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : NULL
##   ..@ vjust        : num 1
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 2.75 0 0 0
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.title.x.top                : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : NULL
##   ..@ vjust        : num 0
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 0 0 2.75 0
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.title.x.bottom             : NULL
##  $ axis.title.y                    : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : NULL
##   ..@ vjust        : num 1
##   ..@ angle        : num 90
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 0 2.75 0 0
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.title.y.left               : NULL
##  $ axis.title.y.right              : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
```

```
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : NULL
##   ..@ vjust        : num 1
##   ..@ angle        : num -90
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 0 0 0 2.75
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.text                       : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : 'rel' num 0.8
##   ..@ hjust        : NULL
##   ..@ vjust        : NULL
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : NULL
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.text.x                     : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : NULL
##   ..@ vjust        : num 1
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 2.2 0 0 0
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.text.x.top                 : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : NULL
##   ..@ vjust        : num 0
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 0 0 2.2 0
```

```
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.text.x.bottom              : NULL
##  $ axis.text.y                     : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : num 1
##   ..@ vjust        : NULL
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 0 2.2 0 0
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.text.y.left                : NULL
##  $ axis.text.y.right               : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : num 0
##   ..@ vjust        : NULL
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 0 0 0 2.2
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.text.theta                 : NULL
##  $ axis.text.r                     : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : num 0.5
##   ..@ vjust        : NULL
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 0 2.2 0 2.2
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.ticks                      : <ggplot2::element_line>
##   ..@ colour       : NULL
##   ..@ linewidth    : NULL
##   ..@ linetype     : NULL
```

```
##   ..@ lineend      : NULL
##   ..@ linejoin     : NULL
##   ..@ arrow        : logi FALSE
##   ..@ arrow.fill   : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.ticks.x                    : NULL
##  $ axis.ticks.x.top                : NULL
##  $ axis.ticks.x.bottom             : NULL
##  $ axis.ticks.y                    : NULL
##  $ axis.ticks.y.left               : NULL
##  $ axis.ticks.y.right              : NULL
##  $ axis.ticks.theta                : NULL
##  $ axis.ticks.r                    : NULL
##  $ axis.minor.ticks.x.top          : NULL
##  $ axis.minor.ticks.x.bottom       : NULL
##  $ axis.minor.ticks.y.left         : NULL
##  $ axis.minor.ticks.y.right        : NULL
##  $ axis.minor.ticks.theta          : NULL
##  $ axis.minor.ticks.r              : NULL
##  $ axis.ticks.length               : 'rel' num 0.5
##  $ axis.ticks.length.x             : NULL
##  $ axis.ticks.length.x.top         : NULL
##  $ axis.ticks.length.x.bottom      : NULL
##  $ axis.ticks.length.y             : NULL
##  $ axis.ticks.length.y.left        : NULL
##  $ axis.ticks.length.y.right       : NULL
##  $ axis.ticks.length.theta         : NULL
##  $ axis.ticks.length.r             : NULL
##  $ axis.minor.ticks.length         : 'rel' num 0.75
##  $ axis.minor.ticks.length.x       : NULL
##  $ axis.minor.ticks.length.x.top   : NULL
##  $ axis.minor.ticks.length.x.bottom: NULL
##  $ axis.minor.ticks.length.y       : NULL
##  $ axis.minor.ticks.length.y.left  : NULL
##  $ axis.minor.ticks.length.y.right : NULL
##  $ axis.minor.ticks.length.theta   : NULL
##  $ axis.minor.ticks.length.r       : NULL
##  $ axis.line                       : <ggplot2::element_line>
##   ..@ colour       : NULL
##   ..@ linewidth    : NULL
##   ..@ linetype     : NULL
##   ..@ lineend      : chr "square"
##   ..@ linejoin     : NULL
##   ..@ arrow        : logi FALSE
##   ..@ arrow.fill   : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.line.x                     : NULL
##  $ axis.line.x.top                 : NULL
##  $ axis.line.x.bottom              : NULL
##  $ axis.line.y                     : NULL
##  $ axis.line.y.left                : NULL
##  $ axis.line.y.right               : NULL
##  $ axis.line.theta                 : NULL
##  $ axis.line.r                     : NULL
```

```
##  $ legend.background               : <ggplot2::element_rect>
##   ..@ fill         : NULL
##   ..@ colour       : logi NA
##   ..@ linewidth    : NULL
##   ..@ linetype     : NULL
##   ..@ linejoin     : NULL
##   ..@ inherit.blank: logi TRUE
##  $ legend.margin                   : NULL
##  $ legend.spacing                  : 'rel' num 2
##  $ legend.spacing.x                : NULL
##  $ legend.spacing.y                : NULL
##  $ legend.key                      : NULL
##  $ legend.key.size                 : 'simpleUnit' num 1.2lines
##   ..- attr(*, "unit")= int 3
##  $ legend.key.height               : NULL
##  $ legend.key.width                : NULL
##  $ legend.key.spacing              : NULL
##  $ legend.key.spacing.x            : NULL
##  $ legend.key.spacing.y            : NULL
##  $ legend.key.justification        : NULL
##  $ legend.frame                    : NULL
##  $ legend.ticks                    : NULL
##  $ legend.ticks.length             : 'rel' num 0.2
##  $ legend.axis.line                : NULL
##  $ legend.text                     : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : 'rel' num 0.8
##   ..@ hjust        : NULL
##   ..@ vjust        : NULL
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : NULL
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ legend.text.position            : NULL
##  $ legend.title                    : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : num 0
##   ..@ vjust        : NULL
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : NULL
##   ..@ debug        : NULL
```

```
##   ..@ inherit.blank: logi TRUE
##  $ legend.title.position           : NULL
##  $ legend.position                 : chr "right"
##  $ legend.position.inside          : NULL
##  $ legend.direction                : NULL
##  $ legend.byrow                    : NULL
##  $ legend.justification            : chr "center"
##  $ legend.justification.top        : NULL
##  $ legend.justification.bottom     : NULL
##  $ legend.justification.left       : NULL
##  $ legend.justification.right      : NULL
##  $ legend.justification.inside     : NULL
##   [list output truncated]
##  @ complete: logi TRUE
##  @ validate: logi TRUE
```

Syntax

Syntax_Mean_TreeDepth (average dependency tree depth)

How often does any subordinate structure appear at all?

```
m0_mtd <- lmer(
  Syntax_Mean_TreeDepth ~ source_is_synthetic * clinical_group +
    factor(round_id) + (1 | cluster_id),
  data = df_4pop,
  REML = FALSE
)

m1_mtd <- lmer(
  Syntax_Mean_TreeDepth ~ source_is_synthetic * clinical_group +
    Age_c + factor(round_id) + (1 | cluster_id),
  data = df_4pop,
  REML = FALSE
)

m2_mtd <- lmer(
  Syntax_Mean_TreeDepth ~ source_is_synthetic * clinical_group +
    Age_c +
    source_is_synthetic:Age_c +
    clinical_group:Age_c +
    factor(round_id) + (1 | cluster_id),
  data = df_4pop,
  REML = FALSE
)

m3_mtd <- lmer(
  Syntax_Mean_TreeDepth ~ source_is_synthetic * clinical_group * Age_c +
    factor(round_id) + (1 | cluster_id),
  data = df_4pop,
  REML = FALSE
)
```

```r
anova(m0_mtd, m1_mtd, m2_mtd, m3_mtd)
```

```
## Data: df_4pop
## Models:
## m0_mtd: Syntax_Mean_TreeDepth ~ source_is_synthetic * clinical_group + factor(round_id) + (1 | clust
## m1_mtd: Syntax_Mean_TreeDepth ~ source_is_synthetic * clinical_group + Age_c + factor(round_id) + (1
## m2_mtd: Syntax_Mean_TreeDepth ~ source_is_synthetic * clinical_group + Age_c + source_is_synthetic:Ag
## m3_mtd: Syntax_Mean_TreeDepth ~ source_is_synthetic * clinical_group * Age_c + factor(round_id) + (1
##        npar     AIC     BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
## m0_mtd    9 -761.24 -725.01 389.62   -779.24
## m1_mtd   10 -762.80 -722.54 391.40   -782.80 3.5575  1    0.05928 .
## m2_mtd   12 -762.33 -714.02 393.16   -786.33 3.5260  2    0.17153
## m3_mtd   13 -761.27 -708.94 393.64   -787.27 0.9460  1    0.33073
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

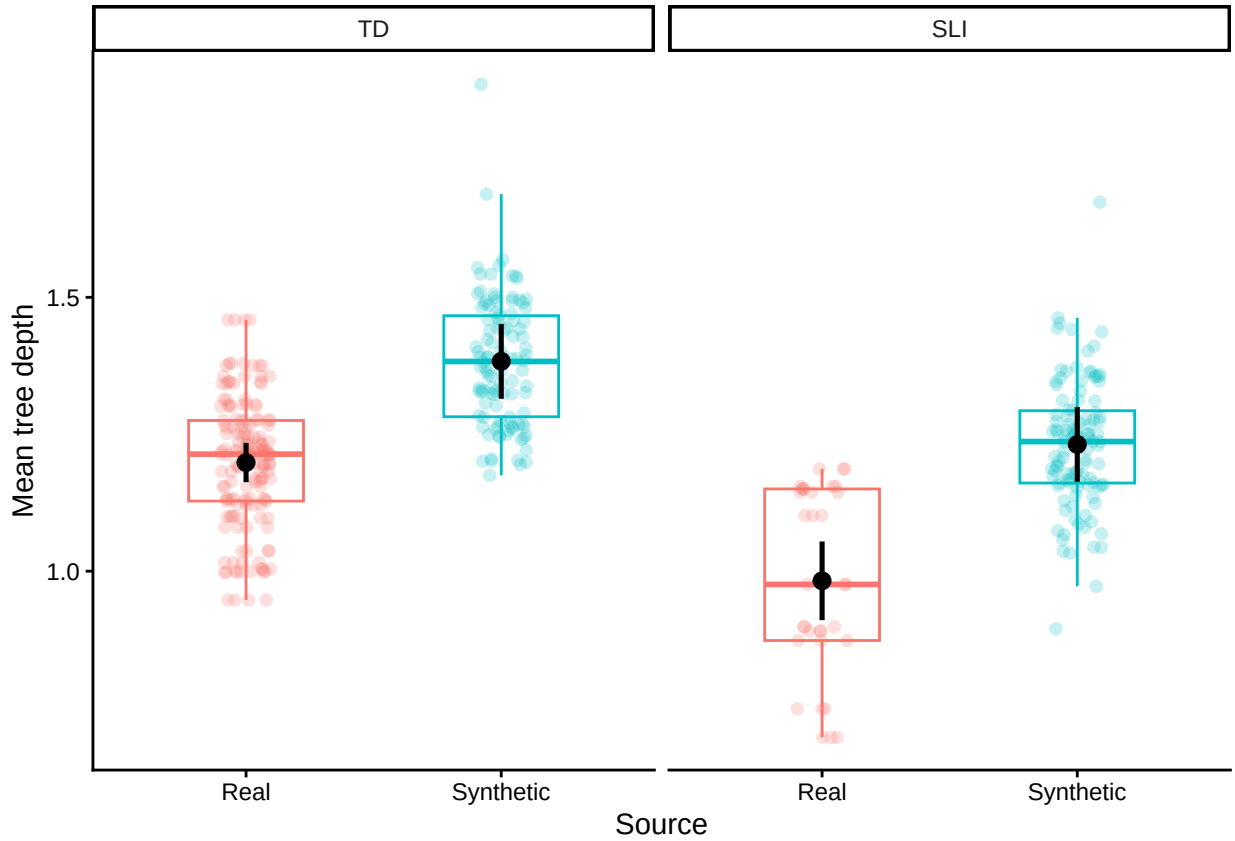
```r
summary(m1_mtd)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula: Syntax_Mean_TreeDepth ~ source_is_synthetic * clinical_group +
##     Age_c + factor(round_id) + (1 | cluster_id)
##    Data: df_4pop
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    -762.8    -722.5     391.4    -782.8       404
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.3951 -0.3173  0.0240  0.3168  6.3604
##
## Random effects:
##  Groups     Name        Variance Std.Dev.
##  cluster_id (Intercept) 0.011325 0.10642
##  Residual               0.006166 0.07852
## Number of obs: 414, groups:  cluster_id, 65
##
## Fixed effects:
##                                          Estimate Std. Error         df t value
## (Intercept)                             1.198e+00  1.997e-02  1.077e+02  59.989
## source_is_synthetic                     1.848e-01  3.727e-02  5.638e+01   4.957
## clinical_groupSLI                      -2.159e-01  3.897e-02  6.752e+01  -5.540
## Age_c                                   1.025e-02  5.375e-03  6.372e+01   1.908
## factor(round_id)round_02                1.206e-03  1.605e-02  3.751e+02   0.075
## factor(round_id)round_03               -9.521e-04  1.605e-02  3.748e+02  -0.059
## factor(round_id)round_04                4.611e-04  1.589e-02  3.694e+02   0.029
## source_is_synthetic:clinical_groupSLI  6.416e-02  4.038e-02  7.778e+01   1.589
##                                        Pr(>|t|)
## (Intercept)                             < 2e-16 ***
## source_is_synthetic                    6.89e-06 ***
## clinical_groupSLI                      5.40e-07 ***
## Age_c                                    0.0609 .
```

```
## factor(round_id)round_02                0.9401
## factor(round_id)round_03                0.9527
## factor(round_id)round_04                0.9769
## source_is_synthetic:clinical_groupSLI   0.1162
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) src_s_ cl_SLI Age_c  f(_)_02 f(_)_03 f(_)_04
## src_s_synth -0.410
## clncl_grSLI -0.387  0.208
## Age_c       -0.007  0.023 -0.008
## fctr(r_)_02 -0.402  0.000  0.000 -0.012
## fctr(r_)_03 -0.401  0.000  0.000 -0.020  0.500
## fctr(r_)_04 -0.404  0.020 -0.004 -0.024  0.505   0.505
## src_s_:_SLI  0.374 -0.238 -0.965  0.003  0.000   0.000   0.004
```

```r
library(dplyr)
library(ggplot2)
library(emmeans)

# Adjusted means from the retained model
emm_mtd <- emmeans(
  m1_mtd,
  ~ source_is_synthetic * clinical_group,
  at = list(source_is_synthetic = c(0, 1))
)

emm_mtd_df <- as.data.frame(emm_mtd) %>%
  mutate(
    source_label = factor(
      source_is_synthetic,
      levels = c(0, 1),
      labels = c("Real", "Synthetic")
    ),
    clinical_group = factor(clinical_group, levels = c("TD", "SLI"))
  )

# Raw data
df_plot <- df_4pop %>%
  mutate(
    source_label = factor(
      source_is_synthetic,
      levels = c(0, 1),
      labels = c("Real", "Synthetic")
    ),
    clinical_group = factor(clinical_group, levels = c("TD", "SLI"))
  ) %>%
  filter(!is.na(Syntax_Mean_TreeDepth))

# Plot
ggplot(df_plot, aes(x = source_label, y = Syntax_Mean_TreeDepth, color = source_label)) +
  geom_boxplot(width = 0.45, alpha = 0.15, outlier.shape = NA) +
  geom_jitter(width = 0.10, alpha = 0.22, size = 1.7) +
```

```
  geom_pointrange(
    data = emm_mtd_df,
    aes(x = source_label, y = emmean, ymin = lower.CL, ymax = upper.CL),
    inherit.aes = FALSE,
    linewidth = 0.9,
    color = "black"
  ) +
  facet_wrap(~ clinical_group) +
  labs(
    x = "Source",
    y = "Mean tree depth"
  ) +
  theme_classic() +
  theme(legend.position = "none")
```

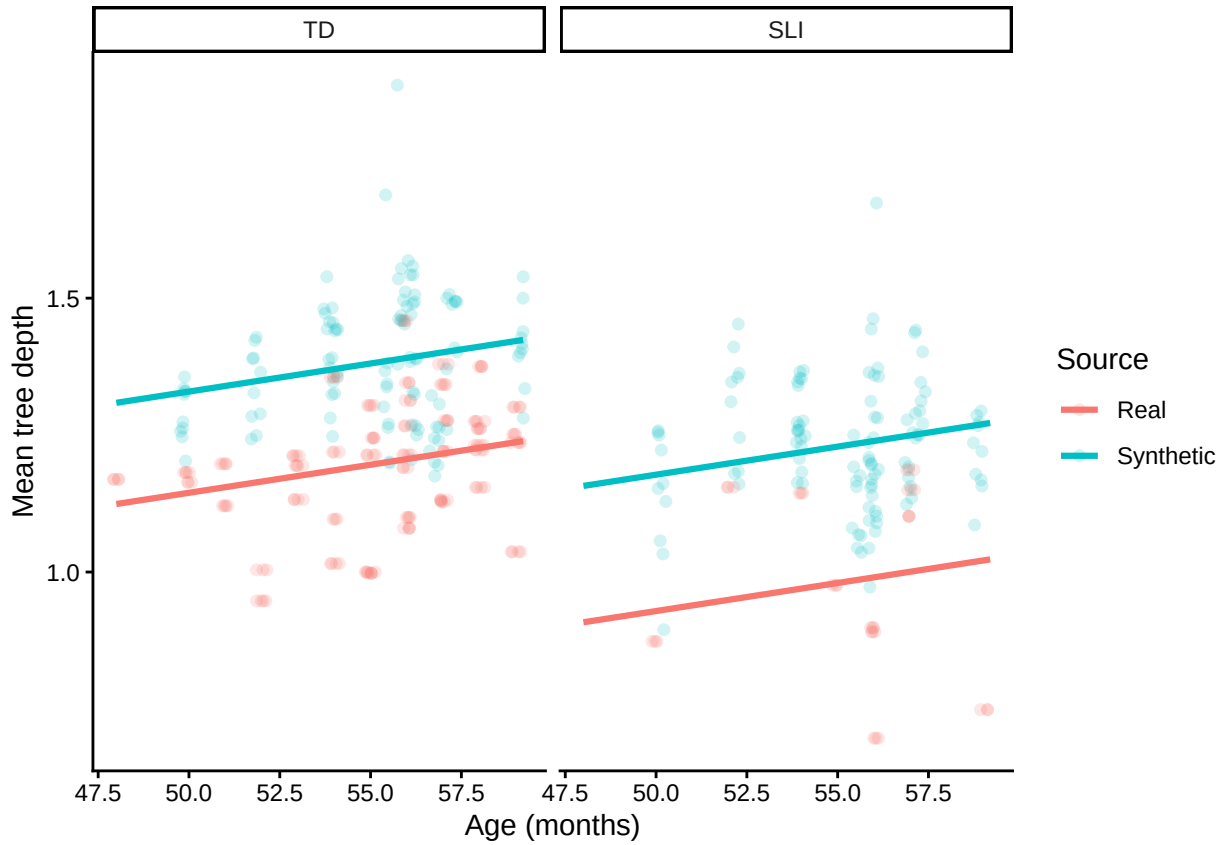


synthetic transcripts have deeper syntactic structure than real. TD > SLI. No apparent difference within sources.

```
library(dplyr)
library(ggplot2)
library(emmeans)

age_mean <- mean(df_4pop$Age_Month, na.rm = TRUE)

df_age_plot <- df_4pop %>%
  mutate(
```

```r
    Age_Month_plot = Age_c + age_mean,
    source_label = factor(
      source_is_synthetic,
      levels = c(0, 1),
      labels = c("Real", "Synthetic")
    ),
    clinical_group = factor(clinical_group, levels = c("TD", "SLI"))
  ) %>%
  filter(
    !is.na(Syntax_Mean_TreeDepth),
    !is.na(Age_c),
    !is.na(Age_Month_plot)
  )

age_grid <- seq(
  min(df_age_plot$Age_c, na.rm = TRUE),
  max(df_age_plot$Age_c, na.rm = TRUE),
  length.out = 100
)

emm_age <- emmeans(
  m1_mtd,
  ~ source_is_synthetic * clinical_group * Age_c,
  at = list(
    source_is_synthetic = c(0, 1),
    Age_c = age_grid
  )
)

pred_df <- as.data.frame(emm_age) %>%
  mutate(
    Age_Month_plot = Age_c + age_mean,
    source_label = factor(
      source_is_synthetic,
      levels = c(0, 1),
      labels = c("Real", "Synthetic")
    ),
    clinical_group = factor(clinical_group, levels = c("TD", "SLI"))
  )

ggplot(df_age_plot, aes(x = Age_Month_plot, y = Syntax_Mean_TreeDepth, color = source_label)) +
  geom_jitter(width = 0.15, height = 0, alpha = 0.18, size = 1.6) +
  geom_line(
    data = pred_df,
    aes(x = Age_Month_plot, y = emmean, color = source_label),
    linewidth = 1.1,
    inherit.aes = FALSE
  ) +
  facet_wrap(~ clinical_group) +
  labs(
    x = "Age (months)",
    y = "Mean tree depth",
    color = "Source"
```

```
  ) +
  theme_classic()
```



Synthetic transcripts appeared to contain greater hierarchical syntactic depth, but this increase in structural complexity was not accompanied by higher developmental morphosyntactic scores; if anything, DSS was slightly lower in the synthetic data.

Syntax_ComplexNP_Utt_Prop

How deep is the dependency structure on average?

```
m0_cnp <- lmer(
  Syntax_ComplexNP_Utt_Prop ~ source_is_synthetic * clinical_group +
    factor(round_id) + (1 | cluster_id),
  data = df_4pop,
  REML = FALSE
)

m1_cnp <- lmer(
  Syntax_ComplexNP_Utt_Prop ~ source_is_synthetic * clinical_group +
    Age_c + factor(round_id) + (1 | cluster_id),
  data = df_4pop,
  REML = FALSE
)
```

```r
m2_cnp <- lmer(
  Syntax_ComplexNP_Utt_Prop ~ source_is_synthetic * clinical_group +
    Age_c +
    source_is_synthetic:Age_c +
    clinical_group:Age_c +
    factor(round_id) + (1 | cluster_id),
  data = df_4pop,
  REML = FALSE
)

m3_cnp <- lmer(
  Syntax_ComplexNP_Utt_Prop ~ source_is_synthetic * clinical_group * Age_c +
    factor(round_id) + (1 | cluster_id),
  data = df_4pop,
  REML = FALSE
)

anova(m0_cnp, m1_cnp, m2_cnp, m3_cnp)
```

```
## Data: df_4pop
## Models:
## m0_cnp: Syntax_ComplexNP_Utt_Prop ~ source_is_synthetic * clinical_group + factor(round_id) + (1 | cl
## m1_cnp: Syntax_ComplexNP_Utt_Prop ~ source_is_synthetic * clinical_group + Age_c + factor(round_id) +
## m2_cnp: Syntax_ComplexNP_Utt_Prop ~ source_is_synthetic * clinical_group + Age_c + source_is_synthetic
## m3_cnp: Syntax_ComplexNP_Utt_Prop ~ source_is_synthetic * clinical_group * Age_c + factor(round_id) +
##        npar     AIC     BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
## m0_cnp    9 -975.85 -939.61 496.92   -993.85
## m1_cnp   10 -974.45 -934.19 497.23   -994.45 0.6041  1     0.4370
## m2_cnp   12 -970.64 -922.33 497.32   -994.64 0.1911  2     0.9089
## m3_cnp   13 -969.18 -916.84 497.59   -995.18 0.5351  1     0.4645
```

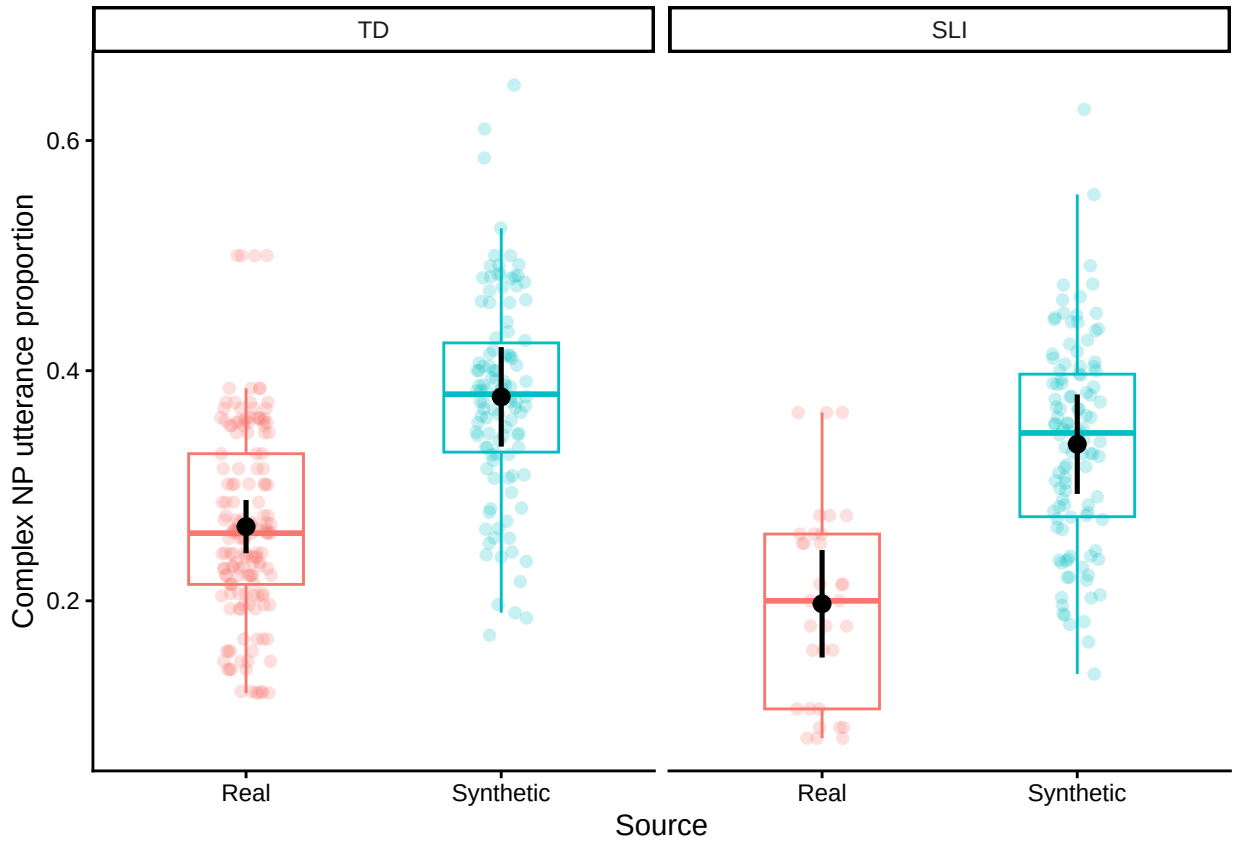
```r
summary(m0_cnp)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula: Syntax_ComplexNP_Utt_Prop ~ source_is_synthetic * clinical_group +
##     factor(round_id) + (1 | cluster_id)
##    Data: df_4pop
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    -975.8    -939.6     496.9    -993.8       405
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.9186 -0.3546 -0.0251  0.3127  3.9660
##
## Random effects:
##  Groups     Name        Variance Std.Dev.
##  cluster_id (Intercept) 0.004436 0.06660
##  Residual               0.003958 0.06291
## Number of obs: 414, groups:  cluster_id, 65
##
```

```
## Fixed effects:
##                                          Estimate Std. Error         df t value
## (Intercept)                             2.645e-01  1.374e-02  1.292e+02  19.256
## source_is_synthetic                     1.128e-01  2.383e-02  5.334e+01   4.732
## clinical_groupSLI                      -6.707e-02  2.552e-02  7.033e+01  -2.628
## factor(round_id)round_02                8.919e-04  1.271e-02  3.902e+02   0.070
## factor(round_id)round_03               -1.590e-03  1.271e-02  3.902e+02  -0.125
## factor(round_id)round_04                5.665e-04  1.261e-02  3.831e+02   0.045
## source_is_synthetic:clinical_groupSLI  2.593e-02  2.689e-02  8.644e+01   0.964
##                                        Pr(>|t|)
## (Intercept)                             < 2e-16 ***
## source_is_synthetic                    1.67e-05 ***
## clinical_groupSLI                        0.0105 *
## factor(round_id)round_02                 0.9441
## factor(round_id)round_03                 0.9005
## factor(round_id)round_04                 0.9642
## source_is_synthetic:clinical_groupSLI    0.3377
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) src_s_ cl_SLI f(_)_02 f(_)_03 f(_)_04
## src_s_synth -0.397
## clncl_grSLI -0.364  0.210
## fctr(r_)_02 -0.462  0.000  0.000
## fctr(r_)_03 -0.462  0.000  0.000  0.500
## fctr(r_)_04 -0.466  0.026 -0.006  0.504   0.504
## src_s_:_SLI  0.345 -0.256 -0.949  0.000   0.000   0.005
```

```r
library(dplyr)
library(ggplot2)
library(emmeans)

# Adjusted means from the retained model
emm_cnp <- emmeans(
  m0_cnp,
  ~ source_is_synthetic * clinical_group,
  at = list(source_is_synthetic = c(0, 1))
)

emm_cnp_df <- as.data.frame(emm_cnp) %>%
  mutate(
    source_label = factor(
      source_is_synthetic,
      levels = c(0, 1),
      labels = c("Real", "Synthetic")
    ),
    clinical_group = factor(clinical_group, levels = c("TD", "SLI"))
  )

# Raw data
df_plot <- df_4pop %>%
  mutate(
    source_label = factor(
```

```r
      source_is_synthetic,
      levels = c(0, 1),
      labels = c("Real", "Synthetic")
    ),
    clinical_group = factor(clinical_group, levels = c("TD", "SLI"))
  ) %>%
  filter(!is.na(Syntax_ComplexNP_Utt_Prop))

# Plot
ggplot(df_plot, aes(x = source_label, y = Syntax_ComplexNP_Utt_Prop, color = source_label)) +
  geom_boxplot(width = 0.45, alpha = 0.15, outlier.shape = NA) +
  geom_jitter(width = 0.10, alpha = 0.22, size = 1.7) +
  geom_pointrange(
    data = emm_cnp_df,
    aes(x = source_label, y = emmean, ymin = lower.CL, ymax = upper.CL),
    inherit.aes = FALSE,
    linewidth = 0.9,
    color = "black"
  ) +
  facet_wrap(~ clinical_group) +
  labs(
    x = "Source",
    y = "Complex NP utterance proportion"
  ) +
  theme_classic() +
  theme(legend.position = "none")
```



synthetic transcripts present higher NP complexity than the real ones.

Syntax_Subordination_Utt_Prop

number of utterances containing subordination / number of analyzed utterances

```
m0_sub <- lmer(
  Syntax_Subordination_Utt_Prop ~ source_is_synthetic * clinical_group +
    factor(round_id) + (1 | cluster_id),
  data = df_4pop,
  REML = FALSE
)

m1_sub <- lmer(
  Syntax_Subordination_Utt_Prop ~ source_is_synthetic * clinical_group +
    Age_c + factor(round_id) + (1 | cluster_id),
  data = df_4pop,
  REML = FALSE
)

m2_sub <- lmer(
  Syntax_Subordination_Utt_Prop ~ source_is_synthetic * clinical_group +
    Age_c +
    source_is_synthetic:Age_c +
    clinical_group:Age_c +
    factor(round_id) + (1 | cluster_id),
```

```r
  data = df_4pop,
  REML = FALSE
)

m3_sub <- lmer(
  Syntax_Subordination_Utt_Prop ~ source_is_synthetic * clinical_group * Age_c +
    factor(round_id) + (1 | cluster_id),
  data = df_4pop,
  REML = FALSE
)

anova(m0_sub, m1_sub, m2_sub, m3_sub)
```

```
## Data: df_4pop
## Models:
## m0_sub: Syntax_Subordination_Utt_Prop ~ source_is_synthetic * clinical_group + factor(round_id) + (1
## m1_sub: Syntax_Subordination_Utt_Prop ~ source_is_synthetic * clinical_group + Age_c + factor(round_
## m2_sub: Syntax_Subordination_Utt_Prop ~ source_is_synthetic * clinical_group + Age_c + source_is_syn
## m3_sub: Syntax_Subordination_Utt_Prop ~ source_is_synthetic * clinical_group * Age_c + factor(round_
##        npar     AIC     BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
## m0_sub    9 -1265.4 -1229.2 641.72   -1283.4
## m1_sub   10 -1269.2 -1228.9 644.59   -1289.2 5.7399  1    0.01658 *
## m2_sub   12 -1267.8 -1219.4 645.87   -1291.8 2.5773  2    0.27565
## m3_sub   13 -1265.9 -1213.5 645.93   -1291.9 0.1193  1    0.72979
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

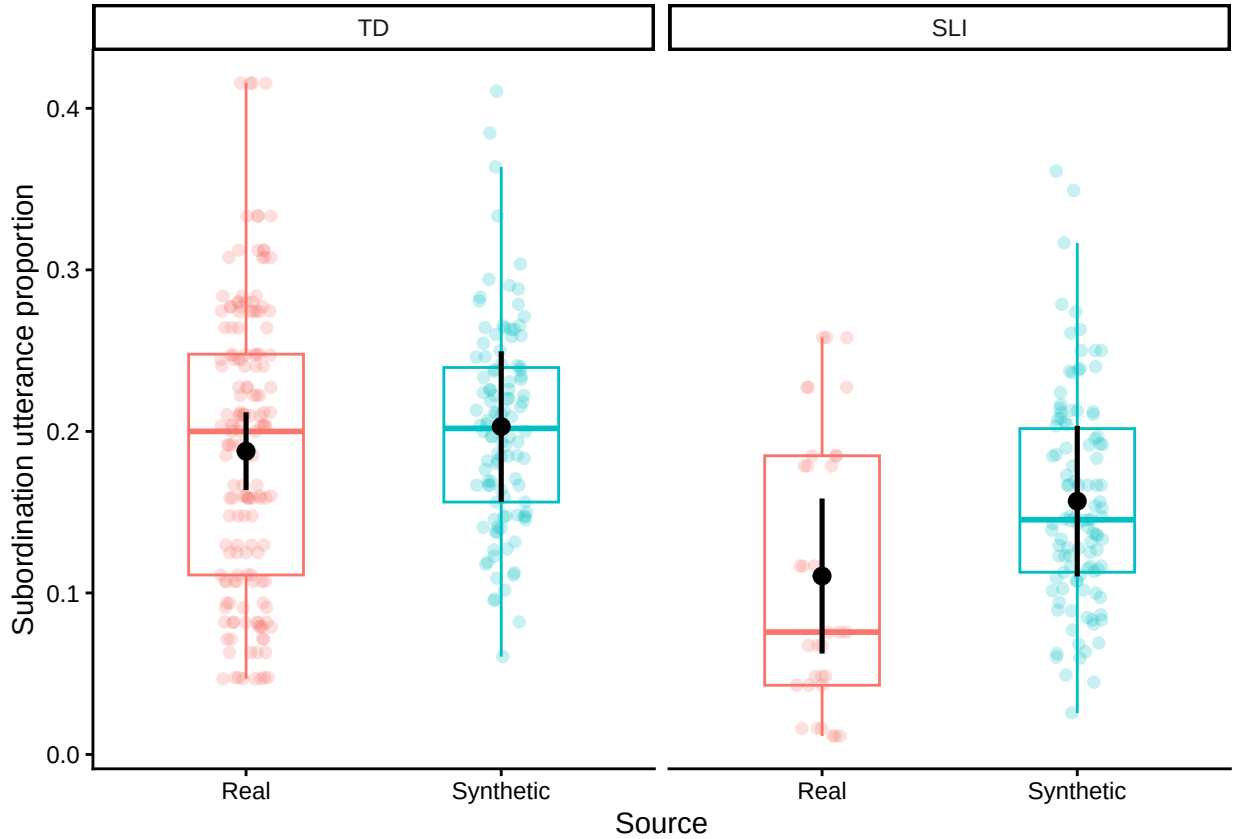
```r
summary(m1_sub)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula:
## Syntax_Subordination_Utt_Prop ~ source_is_synthetic * clinical_group +
##     Age_c + factor(round_id) + (1 | cluster_id)
##    Data: df_4pop
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##   -1269.2   -1228.9     644.6   -1289.2       404
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.4398 -0.2365 -0.0433  0.2359  4.5091
##
## Random effects:
##  Groups     Name        Variance Std.Dev.
##  cluster_id (Intercept) 0.005424 0.07365
##  Residual               0.001673 0.04091
## Number of obs: 414, groups:  cluster_id, 65
##
## Fixed effects:
##                                          Estimate Std. Error         df t value
## (Intercept)                             1.878e-01  1.278e-02  9.133e+01  14.689
## source_is_synthetic                     1.527e-02  2.541e-02  6.006e+01   0.601
```

```
## clinical_groupSLI                     -7.727e-02  2.609e-02  6.682e+01  -2.962
## Age_c                                  8.761e-03  3.613e-03  6.597e+01   2.425
## factor(round_id)round_02               3.171e-04  8.430e-03  3.645e+02   0.038
## factor(round_id)round_03               7.340e-05  8.430e-03  3.644e+02   0.009
## factor(round_id)round_04              -4.739e-04  8.332e-03  3.611e+02  -0.057
## source_is_synthetic:clinical_groupSLI  3.108e-02  2.666e-02  7.290e+01   1.166
##                                       Pr(>|t|)
## (Intercept)                            < 2e-16 ***
## source_is_synthetic                    0.55017
## clinical_groupSLI                      0.00423 **
## Age_c                                  0.01807 *
## factor(round_id)round_02               0.97001
## factor(round_id)round_03               0.99306
## factor(round_id)round_04               0.95467
## source_is_synthetic:clinical_groupSLI  0.24753
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) src_s_ cl_SLI Age_c  f(_)_02 f(_)_03 f(_)_04
## src_s_synth -0.424
## clncl_grSLI -0.409  0.206
## Age_c       -0.013  0.023 -0.008
## fctr(r_)_02 -0.330  0.000  0.000 -0.008
## fctr(r_)_03 -0.330  0.000  0.000 -0.014  0.500
## fctr(r_)_04 -0.332  0.016 -0.003 -0.016  0.506   0.506
## src_s_:_SLI  0.400 -0.224 -0.978  0.003  0.000   0.000   0.003
```

```r
library(dplyr)
library(ggplot2)
library(emmeans)

# Adjusted means from the retained model
emm_sub <- emmeans(
  m1_sub,
  ~ source_is_synthetic * clinical_group,
  at = list(source_is_synthetic = c(0, 1))
)

emm_sub_df <- as.data.frame(emm_sub) %>%
  mutate(
    source_label = factor(
      source_is_synthetic,
      levels = c(0, 1),
      labels = c("Real", "Synthetic")
    ),
    clinical_group = factor(clinical_group, levels = c("TD", "SLI"))
  )

# Raw data
df_plot <- df_4pop %>%
  mutate(
    source_label = factor(
      source_is_synthetic,
```

```r
      levels = c(0, 1),
      labels = c("Real", "Synthetic")
    ),
    clinical_group = factor(clinical_group, levels = c("TD", "SLI"))
  ) %>%
  filter(!is.na(Syntax_Subordination_Utt_Prop))

# Plot
ggplot(df_plot, aes(x = source_label, y = Syntax_Subordination_Utt_Prop, color = source_label)) +
  geom_boxplot(width = 0.45, alpha = 0.15, outlier.shape = NA) +
  geom_jitter(width = 0.10, alpha = 0.22, size = 1.7) +
  geom_pointrange(
    data = emm_sub_df,
    aes(x = source_label, y = emmean, ymin = lower.CL, ymax = upper.CL),
    inherit.aes = FALSE,
    linewidth = 0.9,
    color = "black"
  ) +
  facet_wrap(~ clinical_group) +
  labs(
    x = "Source",
    y = "Subordination utterance proportion"
  ) +
  theme_classic() +
  theme(legend.position = "none")
```



```r
library(emmeans)

pairs(
  emmeans(
    m1_sub,
    ~ source_is_synthetic | clinical_group,
    at = list(source_is_synthetic = c(0, 1))
  ),
  infer = c(TRUE, TRUE),
  adjust = "none"
)
```

```
## clinical_group = TD:
##  contrast                                    estimate     SE   df lower.CL
##  source_is_synthetic0 - source_is_synthetic1  -0.0153 0.0263 65.9  -0.0677
##  upper.CL t.ratio p.value
##    0.0372  -0.581  0.5632
##
## clinical_group = SLI:
##  contrast                                    estimate     SE   df lower.CL
##  source_is_synthetic0 - source_is_synthetic1  -0.0463 0.0335 68.7  -0.1133
##  upper.CL t.ratio p.value
##    0.0206  -1.382  0.1714
##
## Results are averaged over the levels of: round_id
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```